DESIGN OF LOW NOISE LDO WITH HIGH PSR OVER WIDE BANDWIDTH FOR LC VCOs

ECEN 665 FINAL PROJECT REPORT

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OBJECTIVE

To analyze the supply noise effect on the phase noise of the LC VCO and to provide good solution to reduce the pushing factor of LC VCO with good LDO design and other relevant techniques.

ABSTRACT

A Low Noise LDO with high PSR Over wide bandwidth has been designed in 0.5μm technology. This low noise LDO is designed specifically for applications like VCOs and PLLs and the supply noise sensitivity on the phase noise of the LC VCO has been severely reduced using this LDO. Novel nested loop architecture has been used in this LDO design to have better stability. In addition to the robust design of LDO, several other techniques have been explored that can suppress supply noise at higher frequencies at which LDO cannot provide enough power supply rejection. Hence supply noise sensitivity on Phase noise of LC VCO has been reduced drastically over wide range of frequencies.

INTRODUCTION

Voltage Controlled oscillator (VCO) phase-noise performance impacts several RF system specifications such as error-vector-magnitude (EVM), SNR, and SNR under blocking conditions. There are two major sources for VCO phase noise: VCO circuit device noise and ambient noise such as supply, ground, and substrate noise. The noise from supply lines can be classified as:

- Switching noise and interference from other circuit blocks on the same IC, such as dividers and modulators;
- Thermal and noise associated with supply regulation circuits and power supply parasitics.

The supply noise is becoming a critical bottleneck in deep-submicrometer low-noise VCOs due to the reduced power supply voltage, reduced clock swings, higher order of digital integration, and increased number of circuits on the same IC. It has been shown that supply noise has the highest impact on VCO phase noise in terms of deterministic noise [4]. To reduce the effect of supply ripple, low dropout linear regulators are usually preferred for low-noise voltage regulation.

LOW FREQUENCY SUPPLY-NOISE SENSITIVITY OF VCOS

For the analysis of power supply noise on the phase noise of the VCO we have assumed a conventional NMOS cross coupled VCO as shown below.

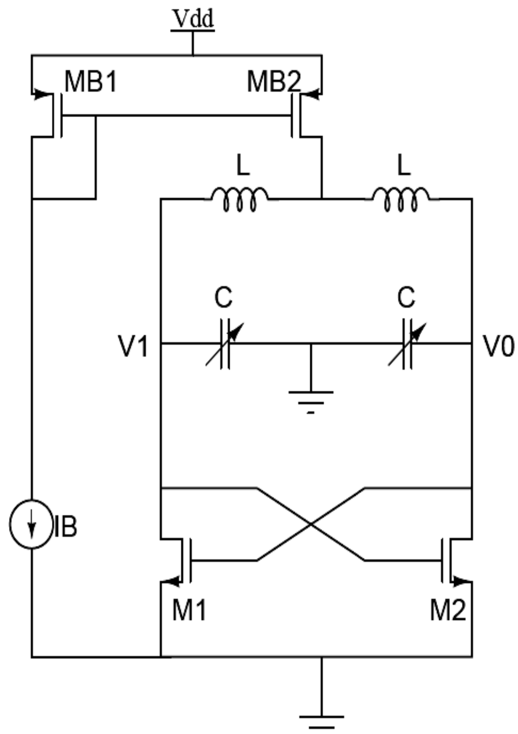


Figure1: NMOS Cross Coupled VCO

Low-frequency power supply noise can be up-converted to the vicinity of the VCO oscillation frequency and therefore contributes to VCO phase noise. This noise conversion is directly related to the power supply frequency pushing factor of the VCO. The power supply pushing factor is defined by the output frequency change corresponding to a change of the power supply voltage while the tuning voltage is kept constant. Due to the bypass capacitance of the local supply regulators, the frequency of interest for phase noise is mainly within the regulator bandwidth. It is a good approximation to assume that the low-frequency power supply noise is constant within one oscillation period.

When the power supply voltage varies by ΔV due to low-frequency power supply noise and ripple, the VCO oscillation frequency will change around the nominal value by $\Delta\omega$ due to a frequency pushing effect, i.e., the VCO oscillates with close-in phase noise. The frequency pushing factor is defined as $= \Delta\omega/\Delta V\ rads/V$.

Representing the low-frequency supply noise by a single-tone $V_n(t) = Vcos(\omega_n t)$, the VCO output signal will be both AM and PM modulated. The VCO output signal can be expressed by

$$V_{out}(t) = A(t)\cos\left[\omega_c t + \theta(\mathrm{t})\right]$$

Where $A(t)$ is the oscillation amplitude, ω_c is the oscillation frequency, and $\theta(t)$ is the oscillation phase.

Assuming a narrowband FM modulation, where $\theta(t) \ll 1$, and assuming a fixed oscillation amplitude, we can approximate the above equation by,

$$V_{out}(t) \approx Acos[\omega_c t + \theta]$$

$$V_{out}(t) = Acos(\omega_c t) - A\theta(t)Sin(\omega_c t)$$

For a given power supply frequency pushing factor k, the instantaneous frequency can be represented by

$$\omega_i(t) = \omega_c + \frac{d\theta(t)}{dt} = \omega_c + kV_n(t)$$

Therefore, instantaneous phase deviation due to supply pushing can be represented by,

$$\theta(t) = \int kV_n(t)dt$$

$$= k \int Vcos(\omega_n t)dt$$

$$= \frac{kV}{\omega_n} Sin(\omega_n t)$$

And the power spectral density of the phase deviation $S_\theta(f)$ can be represented by

$$S_\theta(f) = \frac{1}{4}\left(\frac{\mathrm{kV}}{\omega_\mathrm{n}}\right)^2 [\delta(f - f_n) + \delta(f + f_n)]$$

Representing the amplitude power spectral density of $V_{out}(t)$ by,

$$S_{out}(f) = \frac{A^2}{4}[\delta(f - f_c) + \delta(f + f_c)] + \frac{A^2}{4}[S_\theta(f - f_c) + S_\theta(f + f_c)]$$

We obtain the supply noise source-induced phase noise in dBc/Hz as

$$\mathcal{L}(f_n) = 10\log[S_\theta(f_n)]$$

$$= 10\log\left[\frac{1}{4}\left(\frac{kV}{\omega_n}\right)^2\right]$$

$$= 20\log|k| + 20logV - 20log\omega_n - 6.02$$

This equation shows that the low-frequency power supply induced phase noise magnitude is directly proportional to the supply noise amplitude V and the frequency pushing factor k and inversely proportional to the noise frequency ω_n. The frequency characteristics of the transfer function from low-frequency power supply noise to VCO phase noise is shown in figure below.

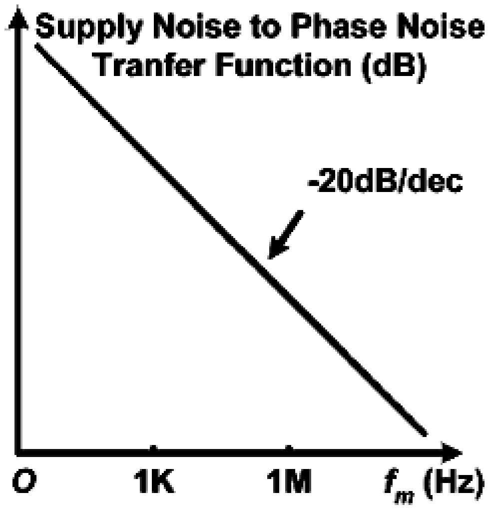


Figure2: Transfer function of low-frequency power supply noise to VCO phase-noise conversion

LC-TANK VCO SUPPLY SENSITIVITY

The power supply noise-induced VCO phase noise can be suppressed by reducing the VCO power supply frequency pushing factor . The VCO phase-noise conversion and frequency pushing have been viewed as AM–PM conversion due to the capacitance nonlinearities in prior literature. In addition to capacitance nonlinearities, the frequency pushing effect also arises from the nonlinearity of the switching devices.

From the analysis of this NMOS cross coupled LC oscillator it can be observed that, during one cycle of operation the transistors spend most of their time in linear region. And the simplified half circuit of the VCO for the switching device in the linear region is shown below.

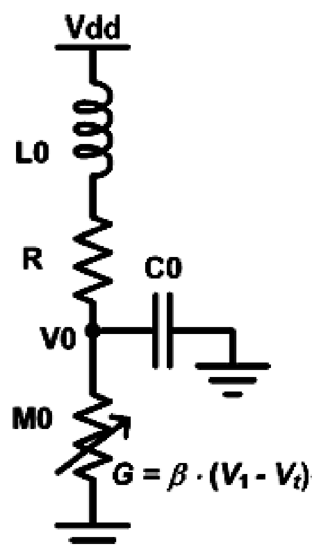


Figure3: simplified half circuit of the VCO for the switching device in the linear region

The resulting half circuit contains a lossy LC tank and the variable resistor representing the switching device in parallel. For a fixed conductance G, the oscillation frequency of the circuit is given by,

$$\omega^2 = \frac{1}{LC} - \left(\frac{RC - GL}{2LC}\right)^2$$

Hence the change of the conductance G will result in a change in oscillation frequency of the RLC tank.

The same relation applies to the variable G . A larger β will result in larger equivalent G. Given that β cannot be too small for starting the oscillation, the increased G will result in lower oscillation frequency. Assuming there is no amplitude control loop, when the power supply voltage V_{dd} increases, the oscillation amplitude increases with it. Therefore, the voltage V_1 between points A and B increases. It results in a bigger equivalent conductance G, resulting in a longer oscillation period. Thus as a first order approximation, if we assume that the equivalent conductance G is proportional to V_{dd}, the pushing factor becomes,

$$k = \frac{\partial\omega}{\partial V_{dd}}\ \alpha\,\frac{\partial\omega}{\partial G} = -\frac{1}{2\omega}.\frac{G}{C}$$

Therefore, the magnitude of the frequency pushing factor k can be reduced by decreasing the equivalent conductance G, which can be achieved by making $\beta = \mu_0 C_{ox} W/L$ small. In addition to this it must be observed that G is also depends on V_{DS} as it is in linear region. Thus any variation in V_{dd} (power supply noise) can be considered as a common mode noise and hence it disturbs the V_{DS} of the transistor.

PHASE NOISE

In explaining the concept of Phase Noise in oscillators in [2], Dr. Ali Hajimiri would have assumed a random current source, $i_n(t)$ whose power spectral density has both a flat region and a 1/f region as shown below. In our case, this random current noise happens to be the power supply noise.

Exploiting the concept of Impulse sensitivity function (ISF), the following equation identifies individual contributions to the total $\phi(t)$ for an arbitrary input current $i(t)$ injected into any circuit nose (in this case the supply node), in terms of the various Fourier coefficients of the ISF.

$$\phi(t) = \frac{1}{q_{max}}\left[C_0 \int_{-\infty}^{t} i(\tau)d\tau + \sum_{n=1}^{\infty} C_n \int_{-\infty}^{t} i(\tau)\cos{(n\omega_0\tau)}d\tau\right]$$

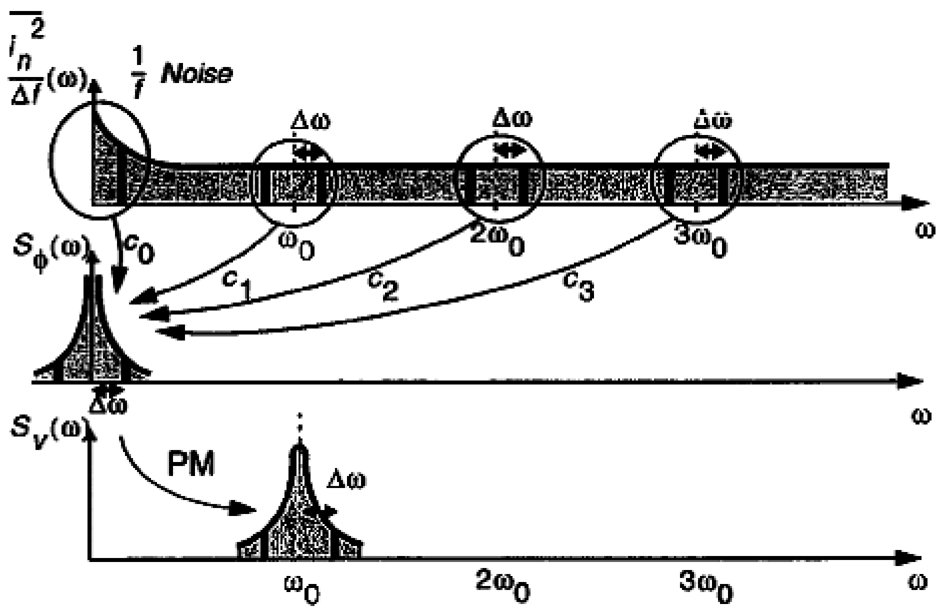


Figure4: Conversion of noise to Phase Fluctuations and Phase-noise sidebands

From the above explained concept it can be understood that there is clear up-conversion and down-conversion of noise from different frequency bands around $n\omega_0$ to ω_0.

This concept of frequency conversion is not observable in ordinary AC simulations. Hence PAC simulations we performed in the NMOS cross coupled VCO to see the frequency up-conversion effects and the results are shown below. It can also be observed that the noise up-conversion takes the shape of Lorentzian pulse.

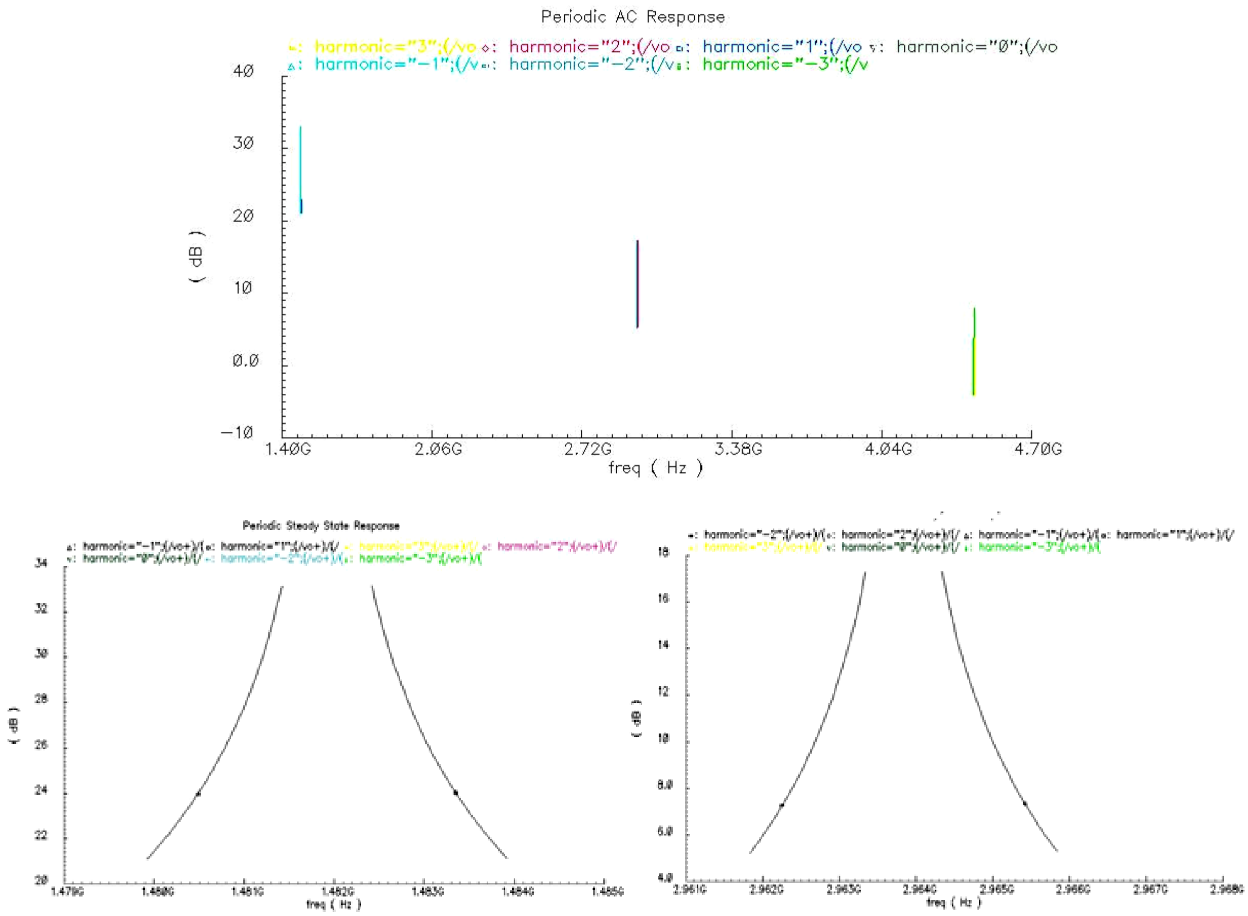


Figure5: Frequency up-conversion of the Supply Noise

Thus to reduce the effect of power supply noise, it can be concluded that the low-frequency noise must be suppressed using good LDO with high PSR over wide bandwidth thus suppressing the amount of low frequency noise being up-converted to ω_0. And the down conversion of high frequency noise (if at all anything is present) to ω_0, must be suppressed using filtering techniques by capacitor proposed above.

UNEXPLORED SOLUTIONS

NOISE FILTERING USING A CAPACITOR

One of the popular technique proposed in [3], utilizes the capacitor C_P to filter out the thermal noise of MB_2 @ $2\omega_0$. This is done primarily to avoided down-conversion of the thermal noise of MB_2 @ $2\omega_0$ to fall around ω_0. In addition to this, it can also be observed that this capacitor helps a lot in filtering out high frequency noise from power supply line. Though low frequency supply noise will be suppressed by the LDO there always exists some challenge in filtering the high frequency supply noise as achieving high PSR for wide range of frequencies is still a potential a research problem till date. To overcome such shortcomings, these kind of techniques stand as a good solution.

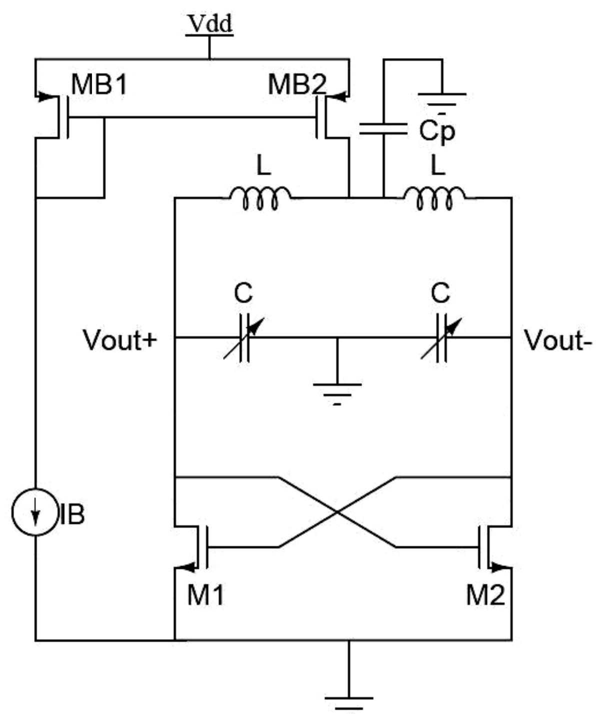


Figure6: Noise filtering using capacitor

This noise filtering technique has resulted in 3dB to 5dB improvement in the case of a noise free V_{DD}. And in the presence of power supply noise, the phase noise improvement is around 8dB, this additional improvement is due to filtering of noise by the capacitor.

NOISE REJECTION USING A CASCODE CURRENT MIRROR

A general and well established concept to improve power supply rejection is to use cascode structure as shown here. Since the impedance offered by the cascode device from V_{dd} to the common mode of the differential circuit is high, the ripples from V_{dd} is very much attenuated. And there is a slight improvement in the phase noise of the VCO.

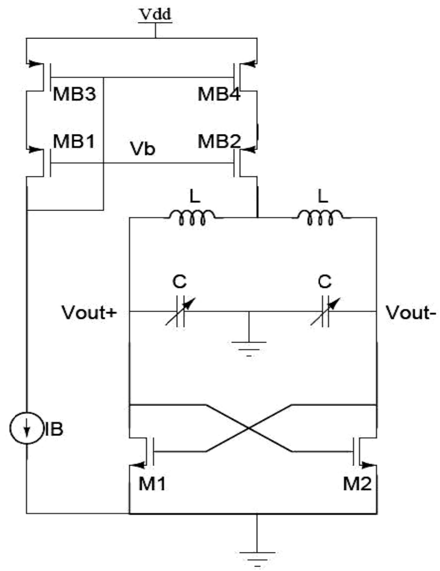


Figure7: Cascoding device to enhance power supply rejection

The improvement in the phase noise by using cascode device is not so drastic. It is roughly around 3 to 4 dB.

The above said concepts can be combined together as shown in the following figure. And the phase noise improvement is more or less the same.

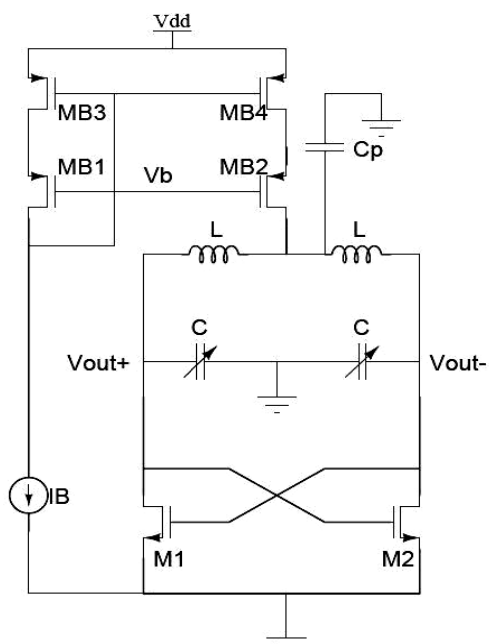


Figure8: Cascoding device and using filtering technique to enhance power supply rejection

EXISTING SOLUTIONS

USING CMFB TO IMPROVE POWER SUPPLY REJECTION IN VCO

Since the power supply noise will occur as common mode noise in output nodes, employing common mode feedback will result in rejecting any variation in the output node's common mode level (the DC level). Hence this would help in rejecting the supply noise coupling to the output nodes. But it must be observed that the CMFB circuit employs active devices and that itself will introduce more noise in the circuit. Hence there isn't that much improvement in Phase noise compared to the other case.

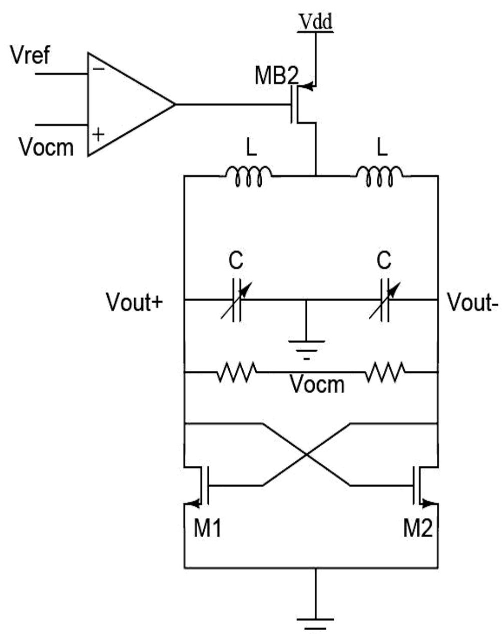


Figure9: Power supply rejection using CMFB at output nodes

USING A VERY GOOD LDO TO REJECT POWER SUPPLY NOISE

The LDO design requirements for the LC VCO is very different from other LDOs used in blocks like baseband (internal digital and analog circuitry), Audio circuits. In order to power radio circuits like VCOs [21], the LDO requirements are as follows.

The above parameters are very essential for any kind of Low Dropout Regulators. Apart from these parameters there are also other important design considerations to be taken care if it is providing supply for VCO & other RF units.

- High PSR over wide range of frequencies
- Very Low Noise Band-gap reference
- LDO itself shouldn't be Noisy

Apart from this, to provide fixed and clean supply voltage even if the following parameters vary,

- Load Current(Load Regulation)
- Input Voltage(Line Regulation)
- Temperature

Low quiescent current and low cost can also be considered.

Since we are dealing with the effect of power noise on the phase noise, our project is mainly focused on improving the PSR for the LDO. Usually the noise from Band Gap is reduced by using a huge off-chip capacitor at its output.

GENERAL LDO BLOCK DIAGRAM

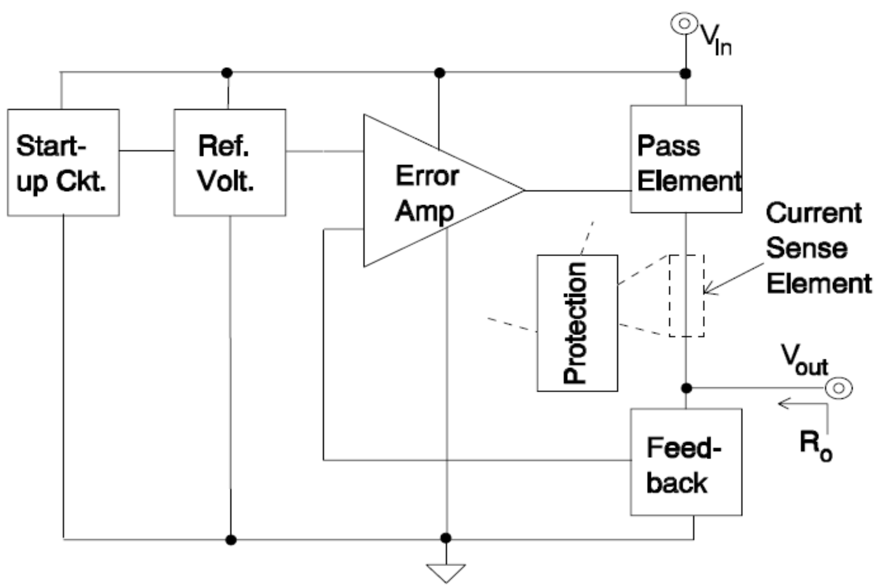
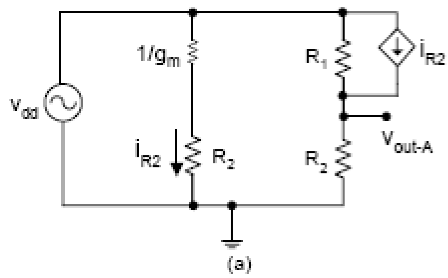


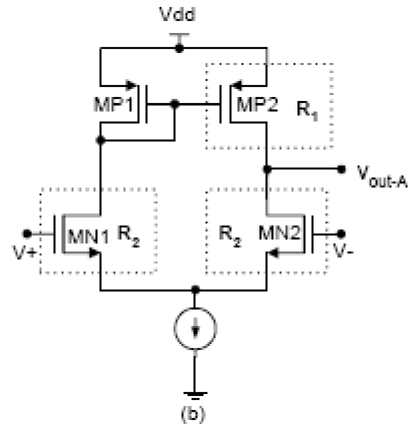
Fig10. LDO Block Diagram

PSR ANALYSIS OF THE NORMAL LDO

PSRR [12]: The error amplifier's output is going to drive the gate of the series pass transistor. The amount of power supply ripples at the output of the error amplifier depends on the type of pass transistor used in LDO.

Case (i) PMOS pass transistor: The source of the pass element is connected to V_{dd}. So if same ripples arrive at the gate then the PSR value will be very high at DC for the LDO since $\Delta V_{sg} \approx 0$. For this case the following error amplifier shown in fig (b) can be used,





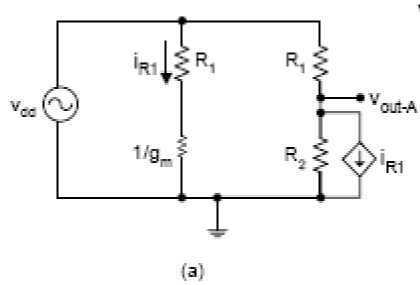
Fig(11a) Small Signal PSR model

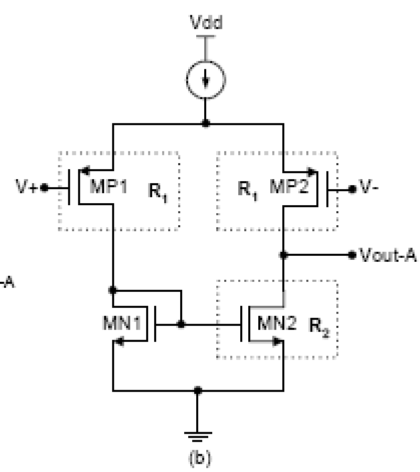
Fig(11b) NMOS Error Amplifier

$$V_{out-A} = \frac{V_{dd}R_2}{R_1 + R_2} + i_{R2}(R_2||R_1) = V_{dd} \quad where\ i_{R2} = \frac{V_{dd}}{R_2}\ (assume \frac{1}{g_m}\ to\ be\ less)$$

Hence, $PSR = \frac{V_{out-A}}{V_{dd}} = 1$ (0 dB)

Case (ii) NMOS Pass transistor: In this case the source of the NMOS is the output of the LDO. So it follows the signal at the gate of the NMOS. Hence very low supply ripples are required at the output of the error amplifier. For this case the following error amplifier shown in fig (b) can be used,





Fig(12a) Small Signal PSR model

Fig(12b) PMOS Error Amplifier

$$V_{out-A} = \frac{V_{dd}R_2}{R_1 + R_2} - i_{R1}(R_1||R_2) = V_{dd} \quad where\ i_{R1} = \frac{V_{dd}}{R_1}\ (assume \frac{1}{g_m}\ to\ be\ less)$$

Hence $PSR = \frac{V_{out-A}}{V_{dd}} = 0$

In NMOS pass transistor case, the dropout voltage is very high compared to that of the PMOS pass transistor case. So in most of the LDO circuits case (i) is preferred.

The above discussion of PSR is for single stage error amplifier only. For more than single stage error amplifier the above discussion need to be extended.

DRAWBACKS WITH NORMAL LDO

The PSR depends on the type of error amplifier used since they are also biased from the same **noisy supply voltage** as that of the pass transistor.

SOLUTION: Is it possible to bias the Error Amplifier and other components except pass transistor with a pure supply (regulated supply)? Let's come to this later.

BW OF THE PSR

Before analyzing the BW for the PSR, the following two types of LDO need to be considered.

i. Internally Compensated LDO (On-Chip Capacitor)
   Dominant pole is within the loop
ii. Externally Compensated LDO (Off-Chip Capacitor)
   Dominant pole is at the output of the LDO

Consider the following simple LDO

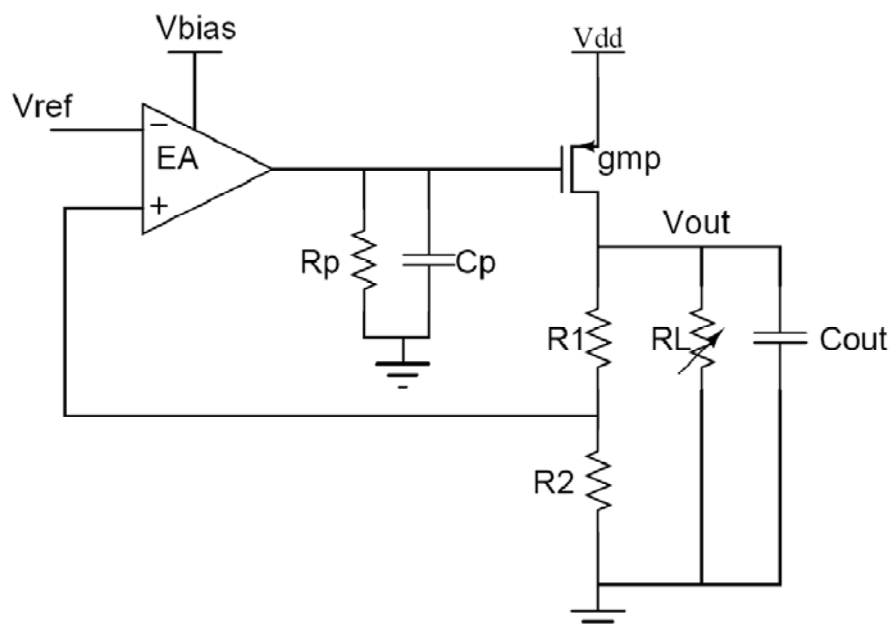


Figure13: Simple LDO architecture

Let's consider the simple block level representation of the LDO for PSR

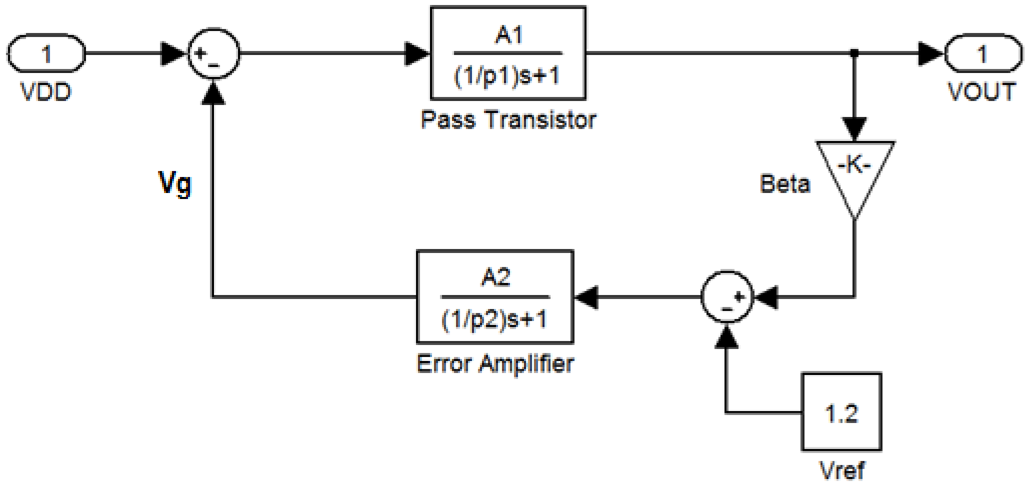


Figure14: MATLAB modeling of the simple LDO

Here Vg is the gate of the pass transistor (PMOS) whose source is connected VDD.

$A_1 = g_{mp}R_{out}$ *, is the gain of the pass transistor*

$P_1 = \dfrac{1}{R_{out}C_{out}}$ *, is the output pole*

$A_2 = g_{mEA}R_P$ *, is the gain from the Error Amplifier*

$P_2 = \dfrac{1}{R_P C_P}$ *, is the pole at the output of the Error Amplifier*

$Beta = \dfrac{R_2}{R_1 + R_2}$

Transfer Function from VDD to VOUT

$$\frac{V_{OUT}}{V_{DD}} = \frac{P_1 P_2 A_1 \left(1 + \frac{s}{P_2}\right)}{s^2 + s(P_1 + P_2) + A_1 A_2 P_1 P_2}$$

Thus the PSR is a band pass plus low pass function

$$\omega_n^2 = A_1 A_2 \beta P_1 P_2 \quad and\ Damping\ Factor\ \zeta = \frac{P_1 + P_2}{2\sqrt{A_1 A_2 \beta P_1 P_2}}$$

CASE (I) INTERNALLY COMPENSATED LDO ($P_2 \ll P_1$)

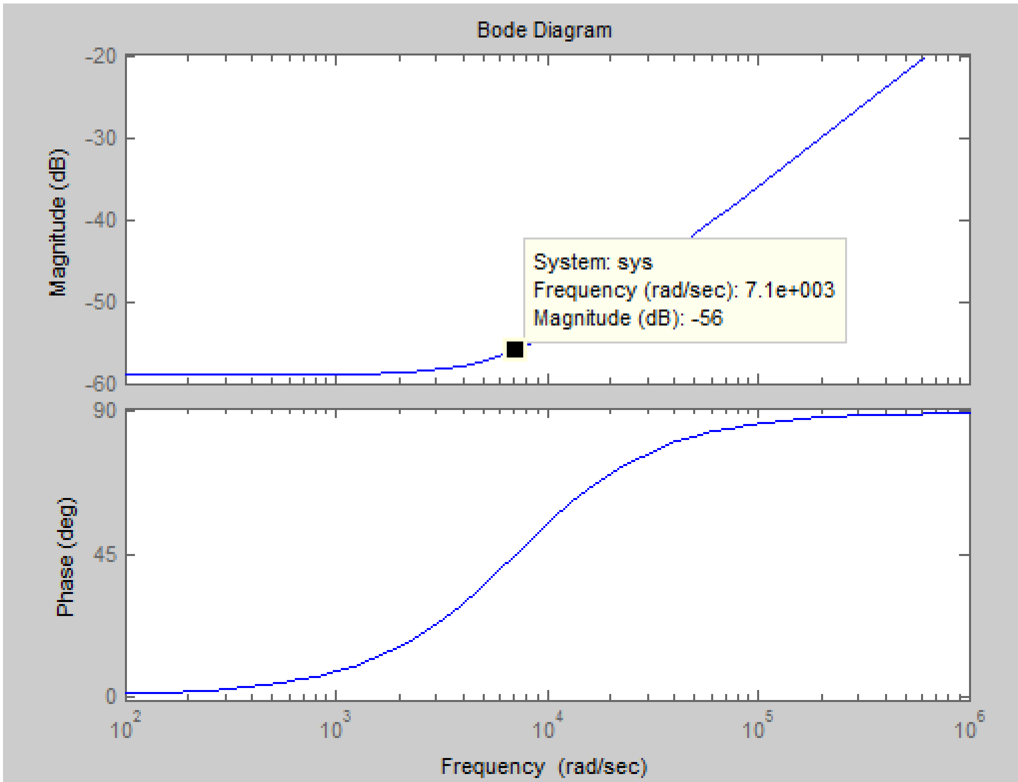


Fig15. PSR of Internally Compensated LDO

This shows for the given specification for the error amplifier, the PSR starts to degrade at $P_2 = 1.13\ KHz$. Hence with internally compensated LDO, the BW of the PSR cannot be increased. In actual PSR expression there will be a pole at the GBW of the open loop transfer function.

The PSR's BW can be increased by increasing the BW of the error amplifier which cannot be increased because of the presence of non-dominant poles. If extra power can be used then higher BW and GBW can be achieved. This means the quiescent current increases which ideally need to be very less in order to increase the efficiency of the LDO.

CASE (II) EXTERNALLY COMPENSATED LDO ($P_1 \ll P_2$)

PSR RESPONSE- UNDER DAMPED

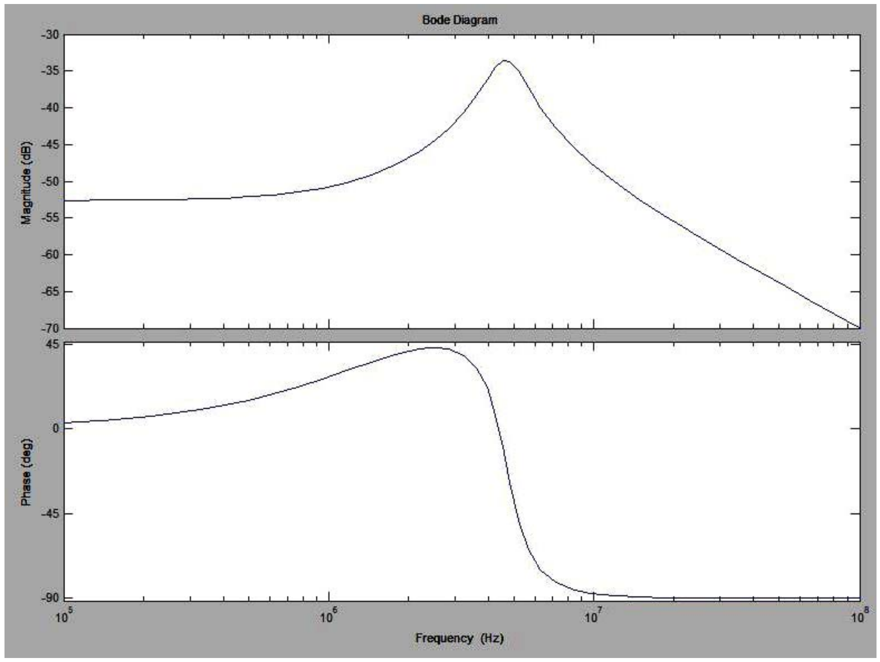


Fig16. $\zeta = 0.171\ (< 0.707)$

PSR RESPONSE - CRITICALLY DAMPED

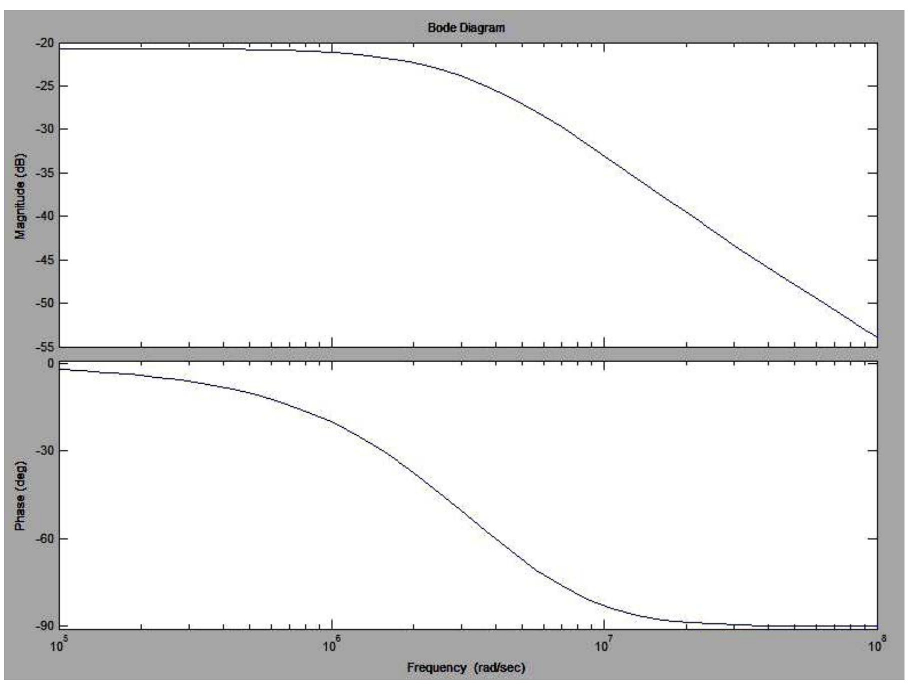


Fig17. $\zeta = 0.707$

Hence based on the damping factor ζ, the PSR response varies. For the under damped case we can achieve a good DC PSR value but the there is a peaking which occurs at ω_n. For the critically damped case there is no peaking but the problem is less DC PSR value.

$$Damping\ Factor\ \zeta = \frac{P_1 + P_2}{2\sqrt{A_1 A_2 \beta P_1 P_2}}$$

Having assumed the pole locations of $P_1\ and\ P_2$ for good stability, the only freedom is to reduce the gains $A_1\ and\ A_2$ to achieve higher ζ value.

Note: In the above modeling it is assumed that the only path from $V_{DD}\ to\ V_{OUT}$ is through pass transistor. But conventional LDO uses Error Amplifier which gets bias from the same noisy V_{DD}.

INFERENCE:

1. There is a tradeoff between peaking and DC PSR value.
2. High PSR over a wide range of frequencies can be achieved only if LDO is externally compensated.
3. Error Amplifier should get bias from regulated supply.

PROPOSED LDO

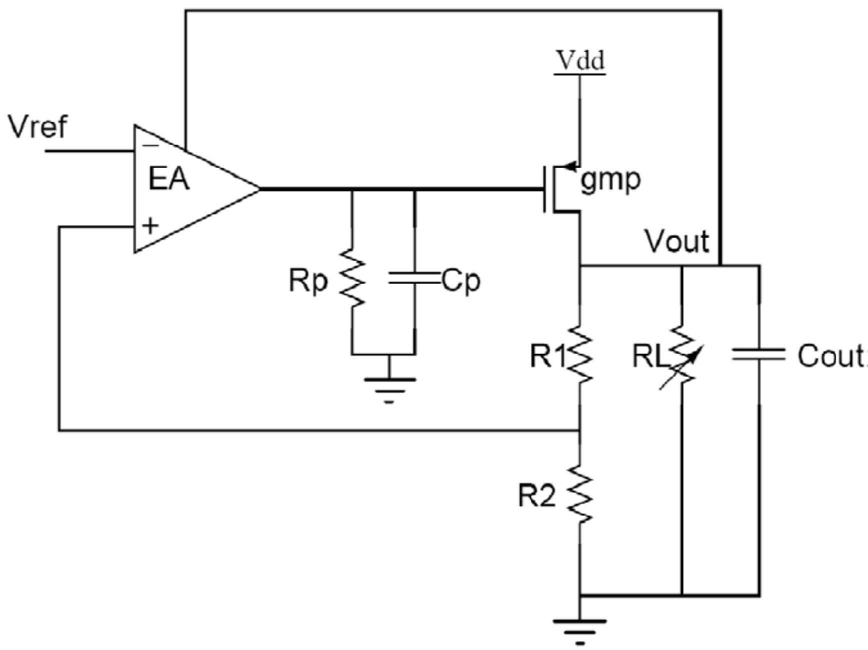


Figure18: Block Level representation of Proposed LDO

V_{OUT} *is used as the supply voltage for the Error Amplifier*

TRANSISTOR LEVEL IMPLEMENTATION

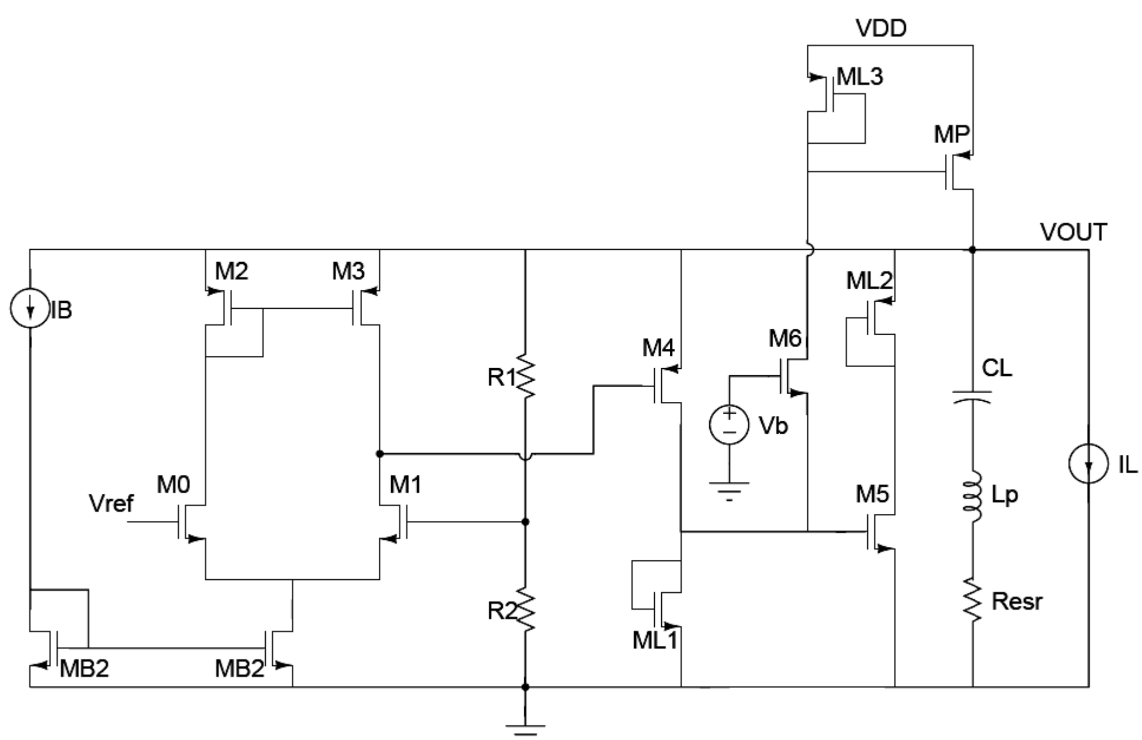


Figure19: Transistor level implementation of Proposed LDO

CONCEPTS BEHIND IMPLEMENTATION

1. **Why dominant pole at the output?**
   a. Noise from LDO will be shaped such that high frequency noise will be better as shown in dotted line in the following figure

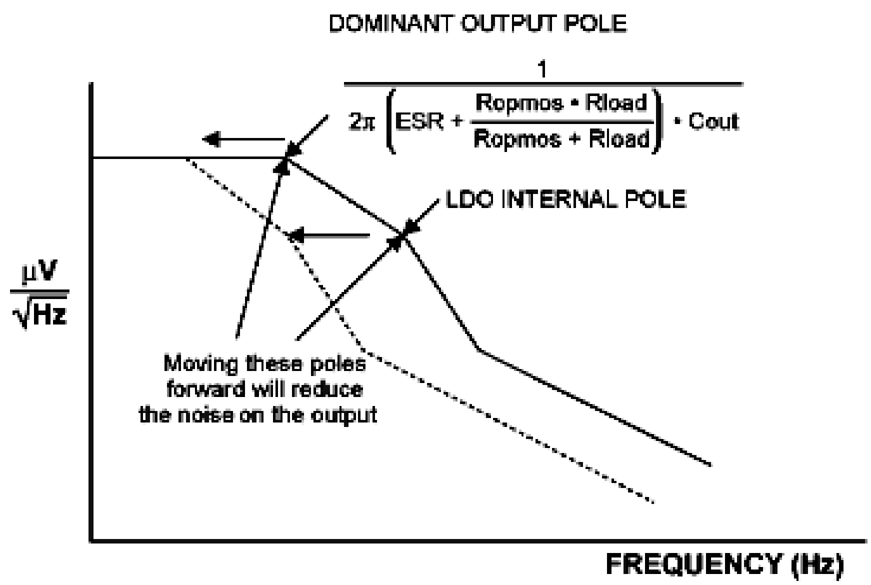


Fig20: Effect of Output Pole Dominant on Noise

   b. High PSR can be achieved over wide range of frequencies while not sparing on the stability.

2. Good stability can be achieved without any compensation scheme or compensating on-chip capacitor. The proposed LDO consists of Nested loops in order to achieve the gain rather cascading the stages which will lead to instability. So each loop can be considered as two stage amplifier and the output pole being the dominant for all loops. This is the **reversed (exact opposite) implementation of Vadim's LDO** for radio units where the dominant pole will always be at the output of first stage (output of EA). The main idea behind our design is to achieve good PSR over wide range of frequencies by having the dominant pole always at the regulated output for the reasons mentioned in the previous discussions on PSR.

   Reference: Vadim Ivanov, Texas Instruments ,A Chapter on "Design methodology and circuit techniques for any-load stable LDOs with instant load regulation and low noise, in Analog Circuit Design, Michiel Steyaert, Arthur H. M. van Roermund and Herman Casier, September 2008.

WORKING PRINCIPLE OF THE PROPOSED LDO

Dominant pole is always at the V_{OUT} node. There are basically three nested loops in the LDO. Transistors $M_0\ to\ M_3$ form the simple Error Amplifier (Differential Pair). $R_1\ and\ R_2$ forms the resistive feedback network. In the first loop M_1 compares the feedback signal with V_{ref} and act as common source stage which is followed by the source follower M_4 to V_{OUT}. The fast loop responding to change in V_{OUT} is through M_4 acting as common gate stage followed by common source stage M_5. For example, if V_{OUT} drops because of sudden increase in the load current, source of M_4 drops and hence the drain of M_4. Thus the drain of M_5 increases since it's V_{gs} has dropped. Vice versa will happen for decrease in load current. In the mean time the third loop formed by common gate (M_4 & M_6) together followed by common source M_P (pass transistor) will help regulation. The first loop helps for low frequency load regulation.

M_6 also helps as a start up transistor. Assume initially there is no voltage at the gate of the pass transistor M_6 and V_{OUT} is at undersired usually zero voltage. Now the source of the M_6 is connected to ground through the diode connected transistor M_{L1}. Hence based on the gate voltage of M_6 sufficient current will flow through the resistor (diode connected) M_{L2} and sufficient gate bias will be generated for the pass transistor which pulls the V_{OUT} to required voltage. $C_L, L_P\ and\ R_{ESR}$ forms the model for the off-chip capacitor (Values are obtained through KEMET Spice)

DESIGN PROCEDURE

The main design consideration is the dominant pole should be at the output of the LDO. Hence huge capacitor of value $1\mu F$ is chosen.
The pass transistor is designed to support the maximum load current and $V_{DSat} < 200\ mV$

$$M_P = \left(\frac{W}{L}\right)_P = \frac{2 * I_{Load}}{\mu_P C_{OX} V_{DSat}^2}$$

The tail current of Error Amplifier is take to be $5\mu A$ and in the resistive feedback network too.

$$R_1 + R_2 = \frac{V_{OUT}}{5\mu A} = 560\ K\Omega$$

V_{ref} is assumed to be 1.2 V.
Thus $\frac{R_2}{R_1+R_2} = 0.4285$
Hence $R_1 = 320\ K\Omega\ \ and\ \ R_2 = 240\ k\Omega$

M_6 transistor sizes are determined in such a way that it helps to bias the pass transistor initially. All the other transistors are sized such that they operate in saturation region. The biasing circuit is shown below ($R_B = 150\ K\Omega$)

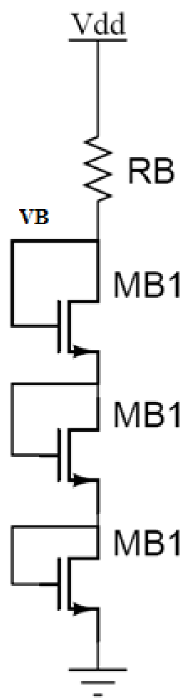


Figure21: Biasing circuit for M_6

SUMMARY OF TRANSISTOR SIZES

Transistor	W (μm)	L (μm)	Multipliers
$M_{0,1}$	4	0.6	2
$M_{2,3}$	4	0.6	2
M_{B1}	7.5	0.6	2
M_{B2}	2	1.2	4
M_4	3	0.6	16
M_5	1.5	1.8	2
M_6	16	0.6	2
M_{L1}	1.5	3	2
M_{L2}	2	1.2	2
M_{L3}	2	0.6	4
M_P	100	0.6	200

Table1: Transistor Sizes

SIMULATION PLOTS

TRANSIENT

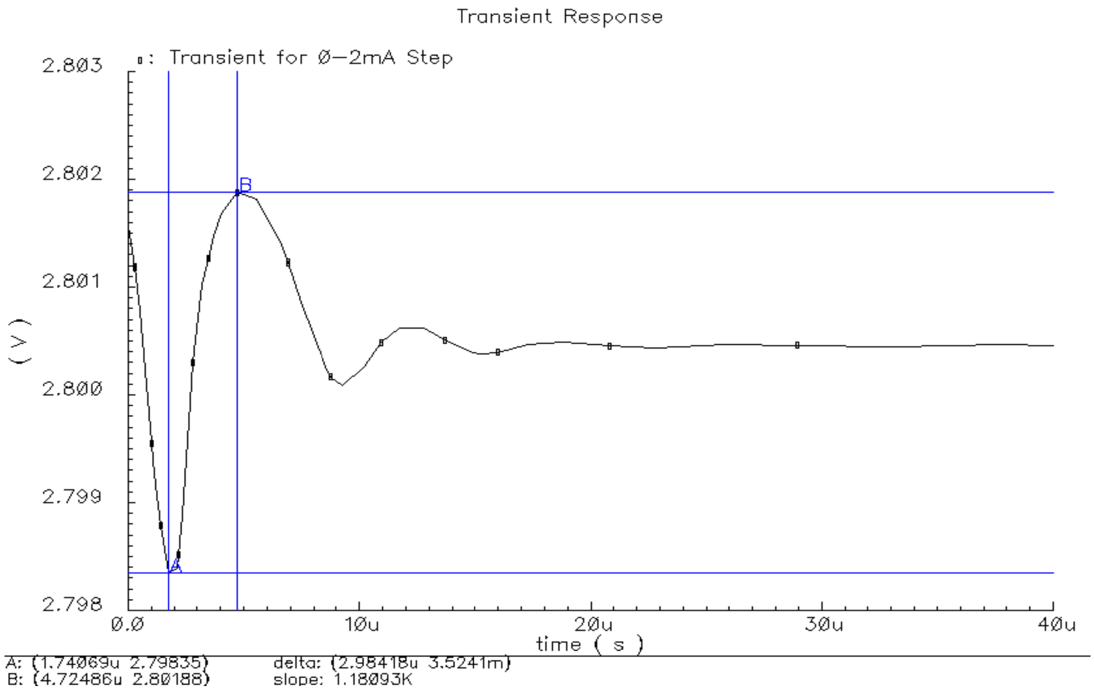


Fig22. Transient for 0-2mA Load Current Step

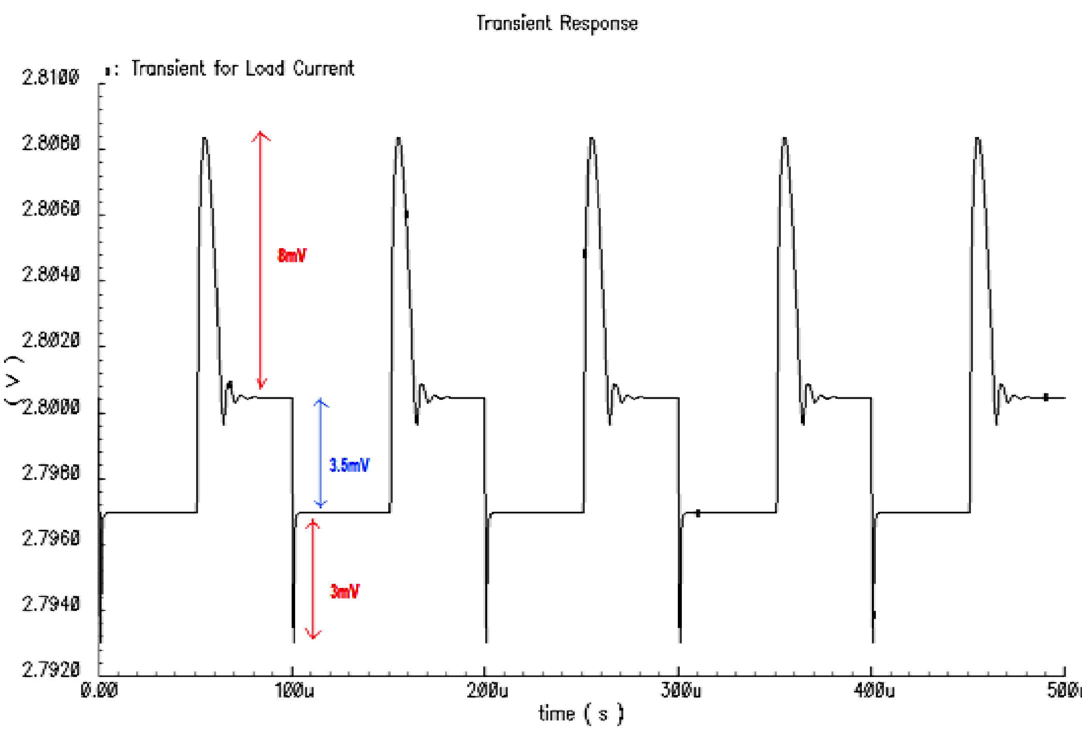


Fig23. Transient for 1mA-20 mA Load Current Step

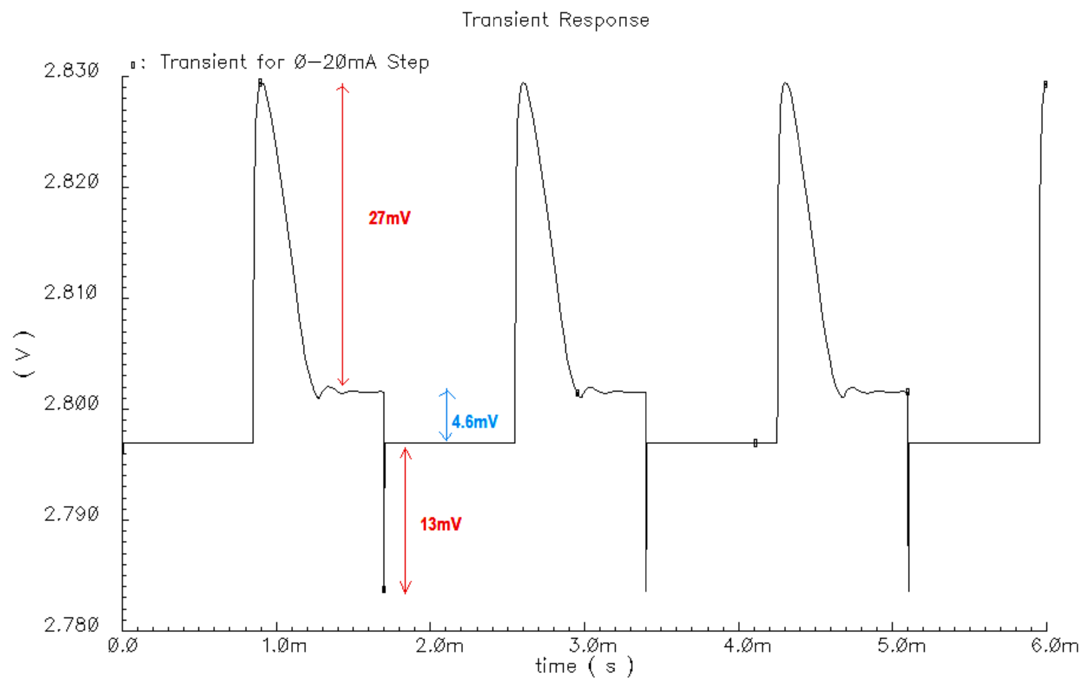


Fig24. Transient for 0-20 mA Load Current Step

AC RESPONSE

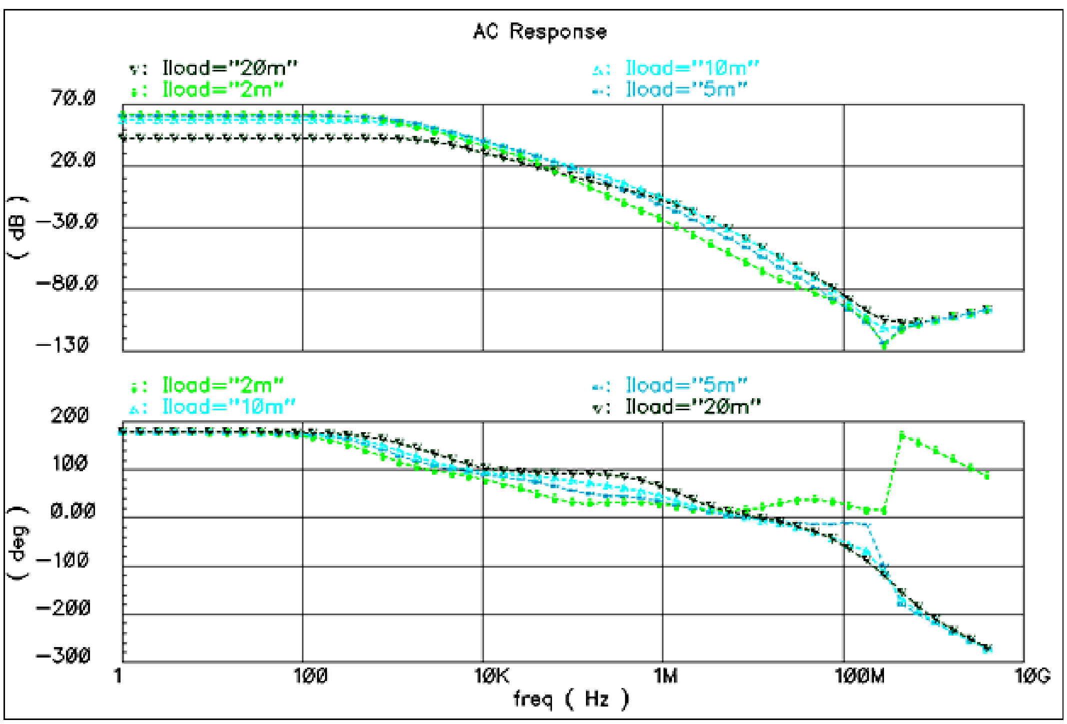


Fig25. AC Response for different Load Currents

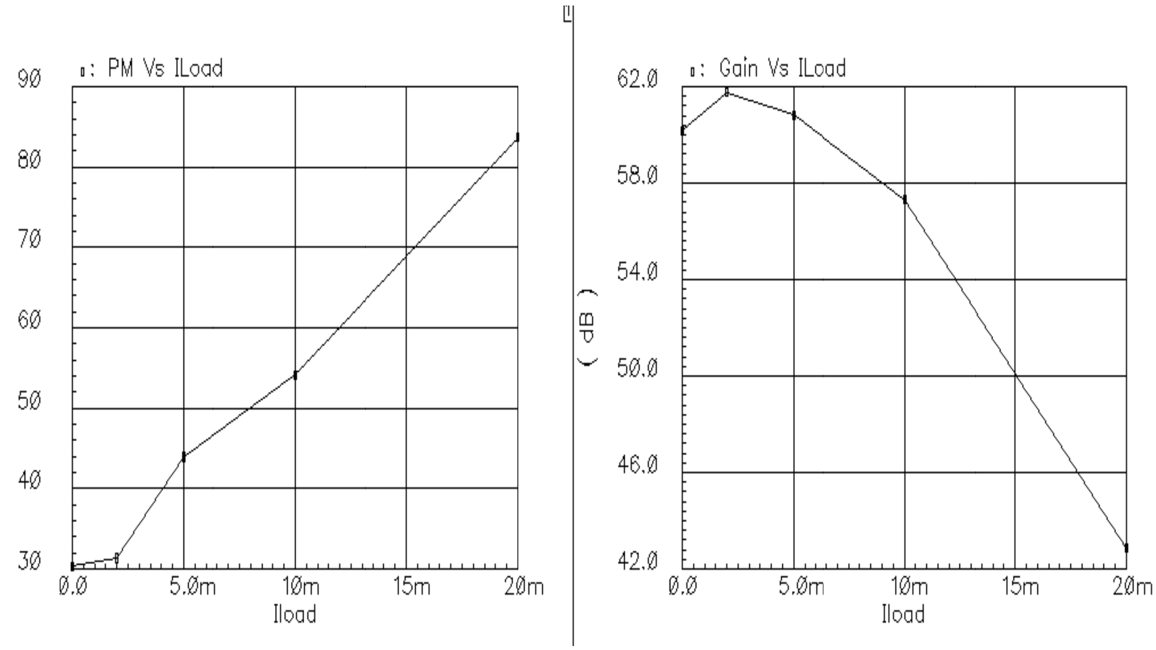


Fig26. Phase Margin and Gain for different Load Currents

At increasing load current stability increases because the pole at the gate of the pass transistor is moving towards higher frequencies but gain decreases since the current through M_{L2} increases.

PSR

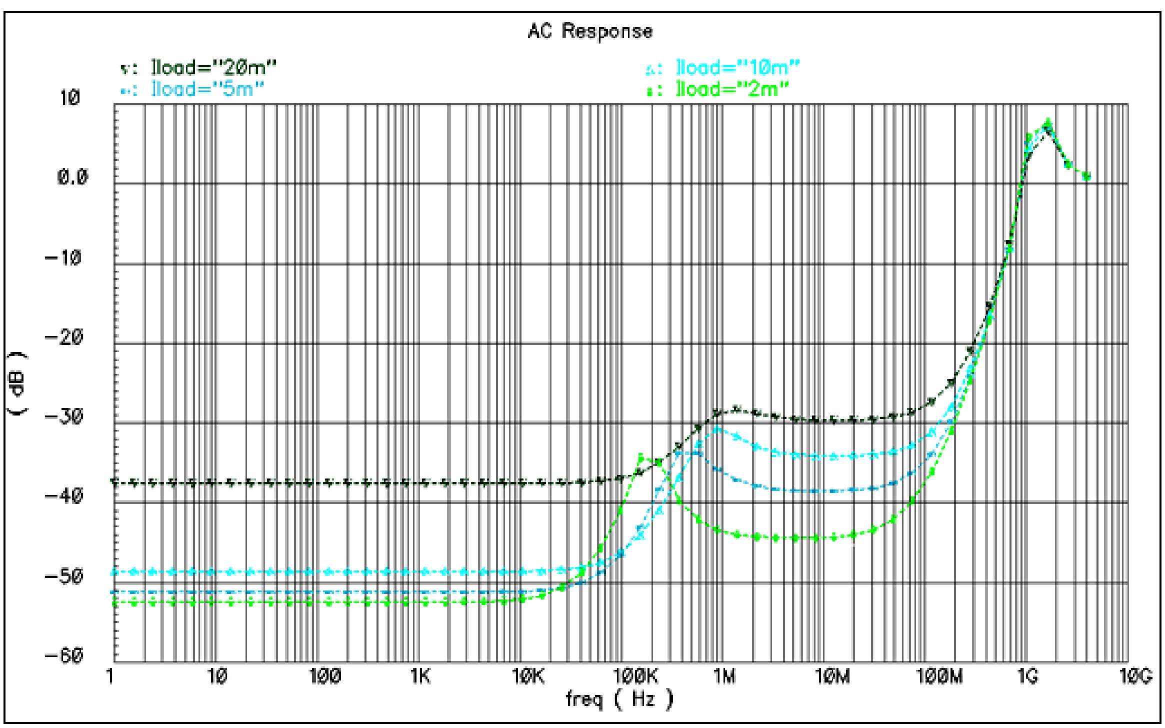


Fig27. PSR for different Load Currents with Real World Off Chip capacitor (including inductor L_P)

LINE REGULATION

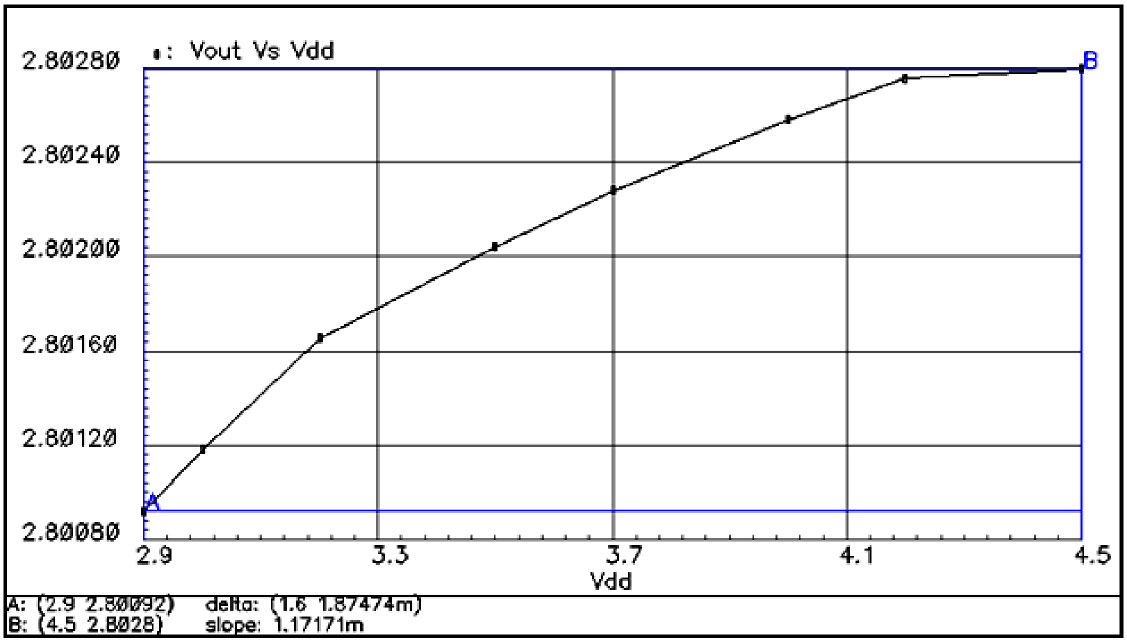


Fig28. Line Regulation

Temperature Coefficient

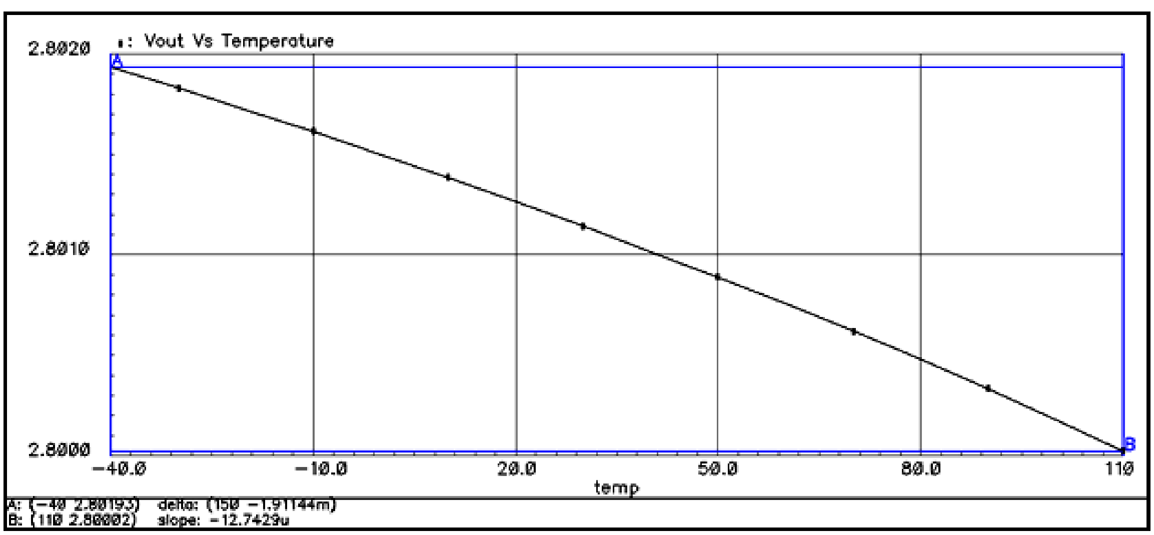


Fig29. Temperature Coefficient

Temperature Coefficient = $\frac{\left(V_{ref}(T_2) - V_{ref}(T_1)\right)}{V_{ref}(27°C) * (T_2 - T_1)}$ in ppm

$V_{ref}(27°C) = 2.80118$

Hence from the curve Temperature Coefficient = 4.55 ppm

Noise

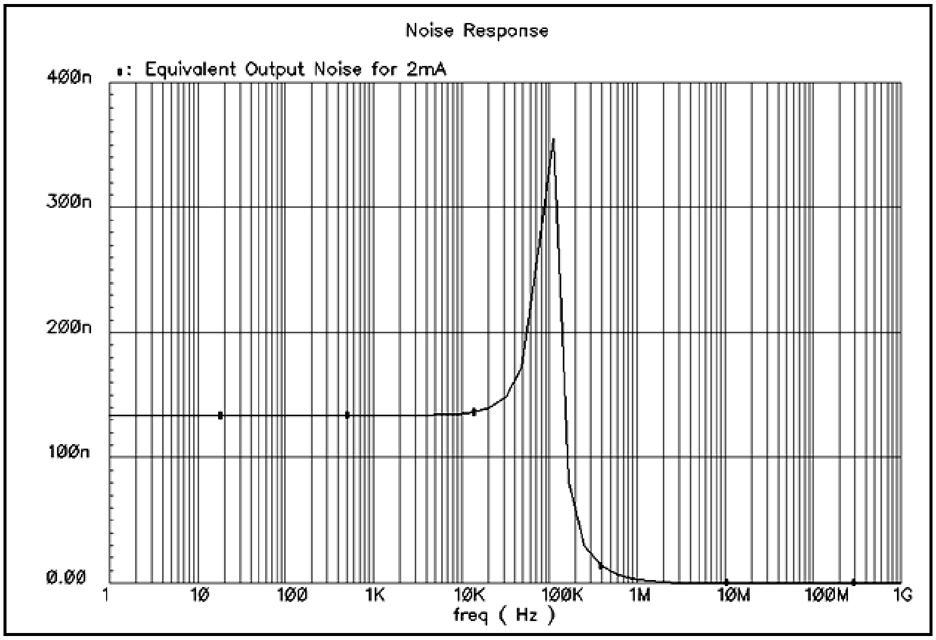


Fig30. Equivalent Output Noise for 2mA Load Current

Typical Simulation Results

PSR Summary

For Real Capacitor

ILOAD(A)	DC	1 MHz	10 MHz	50 MHz	75 MHz	100 MHz	150 MHz
2m	-52.4	-43.5	-44.35	-42	-39.72	-37.5	-33.4
5m	-51.2	-36.05	-38.5	-37.48	-36.3	-34.9	-31.8
10m	-48.6	-30.84	-34.1	-33.5	-32.7	-31.51	-29.5
20m	-37.6	-28.73	-29.6	-29.2	-28.6	-27.86	-26.1

Table2. PSR at various frequencies for different Load Currents including L_P

All the above values are in dB.

Inference:

1. The use of real capacitor with inductor L_P affects the PSR only at higher frequencies.
2. There is a peaking as expected at ω_n and because of zero due to the ESR the PSR goes up as frequency increases.
3. There is just 2 dB to 7 dB variation in the PSR between 10 MHz and 100 MHz

WORST CASE CORNER (PROCESS) VARIATIONS

Four different model files, fast-fast, slow-slow, fast-slow and slow-fast are created from the actual ami06.m file. The variations are accounted according to the MOSIS spice parameters. It has been found that the worst case stability and PSR occurs from slow-fast corner at $110° \, C$.

WORST CASE CORNER (SLOW-FAST) SIMULATION PLOTS

AC RESPONSE

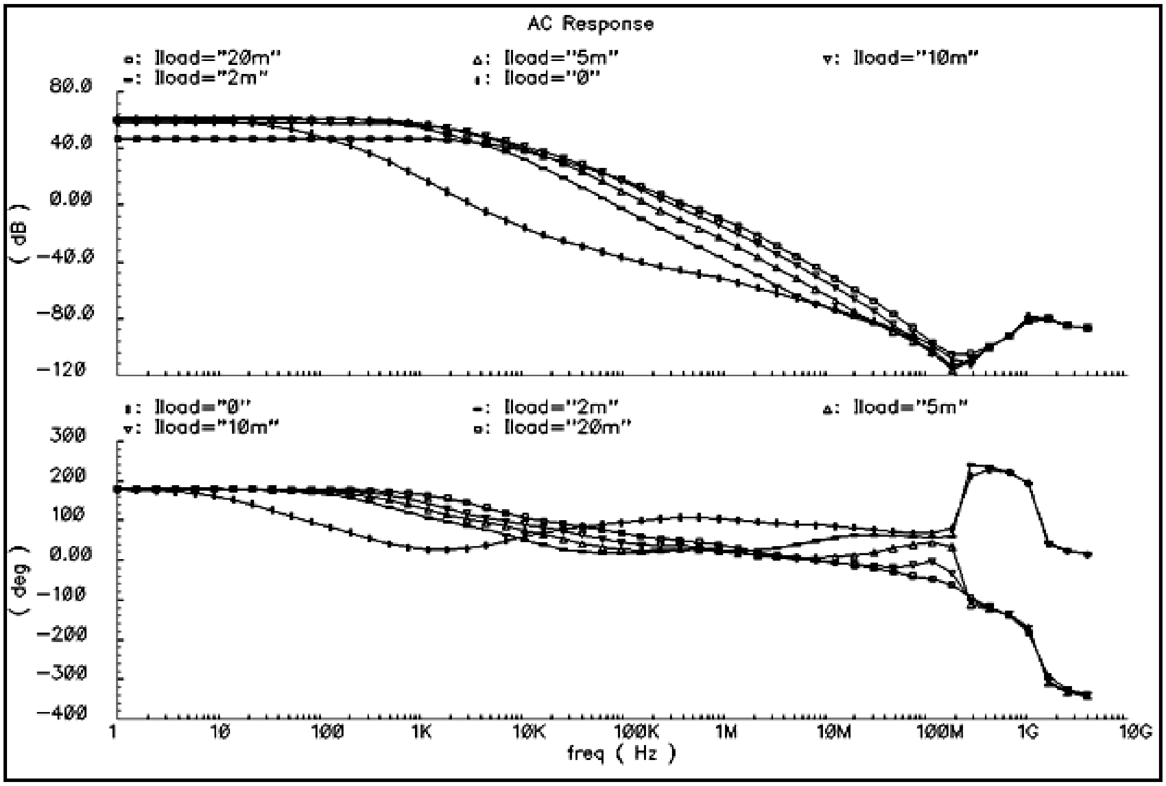


Fig31. Worst Case Frequency Response of the Open Loop

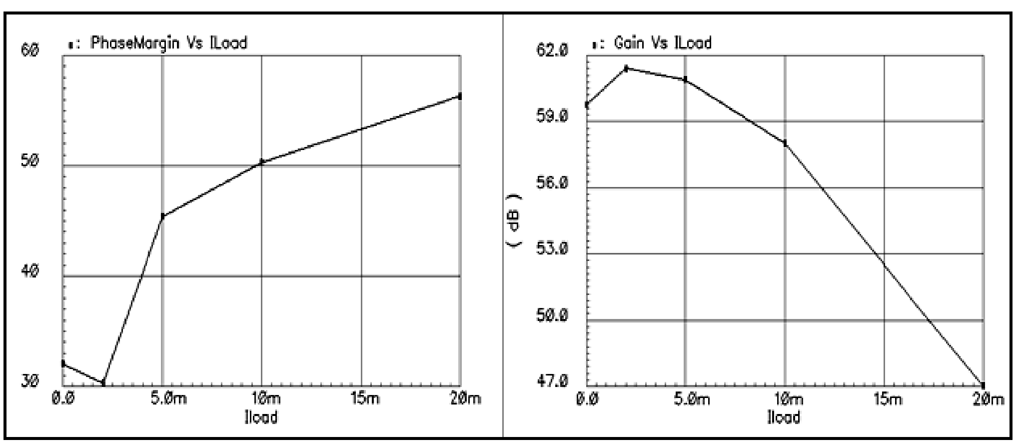


Fig32. Worst Case Phase Margin and Gain Vs ILoad

TRANSIENT RESPONSE

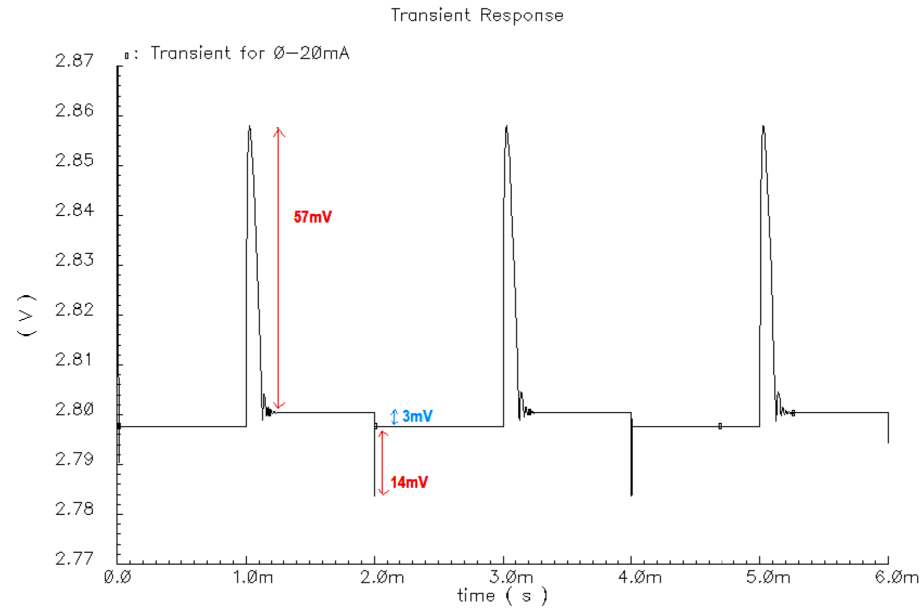


Fig33. Worst Case Transient for 0-20mA Load Current Step

PSR

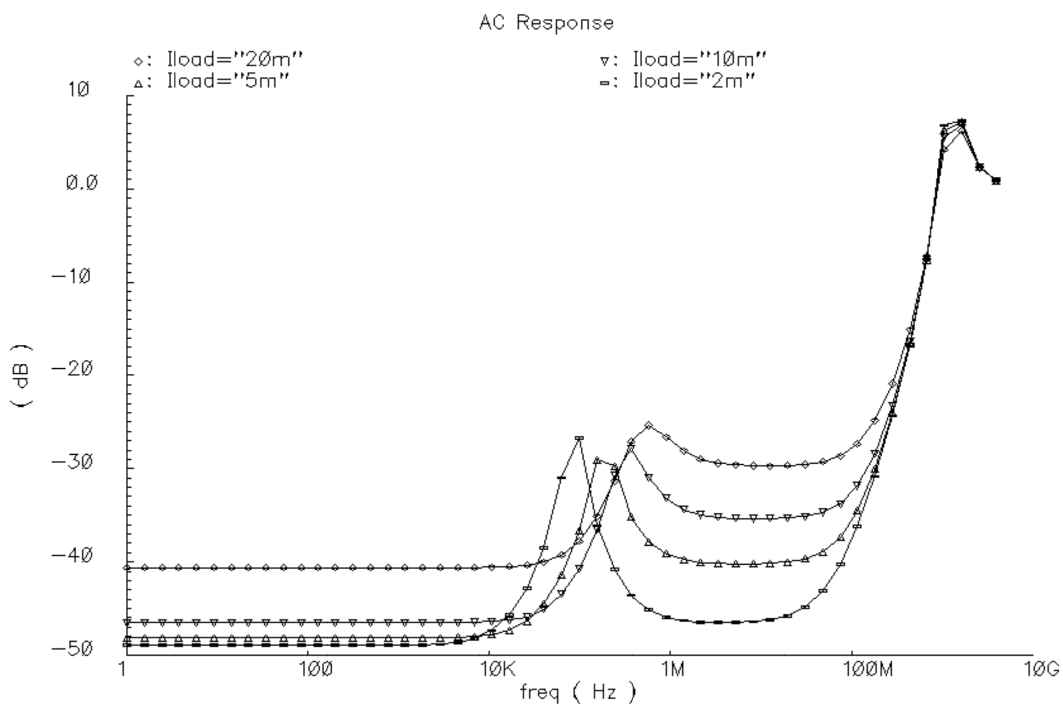


Fig34. Worst Case PSR for different Load Currents

NOISE

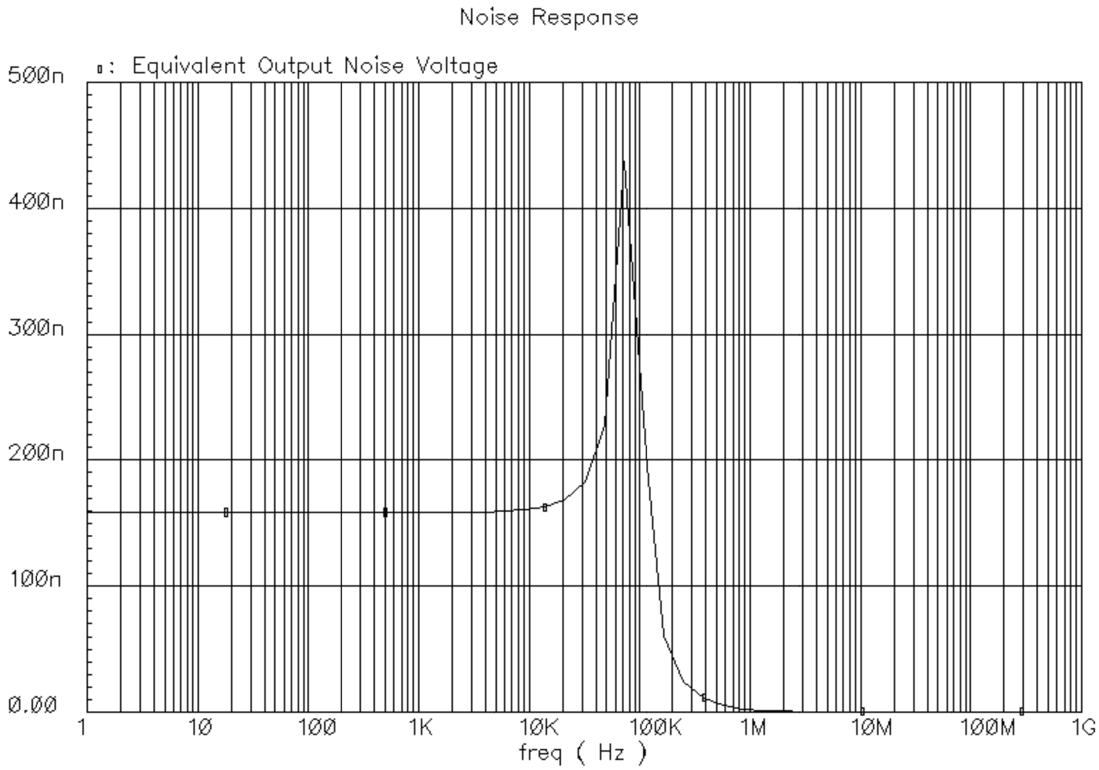


Fig35. Worst Case Noise

LINE REGULATION

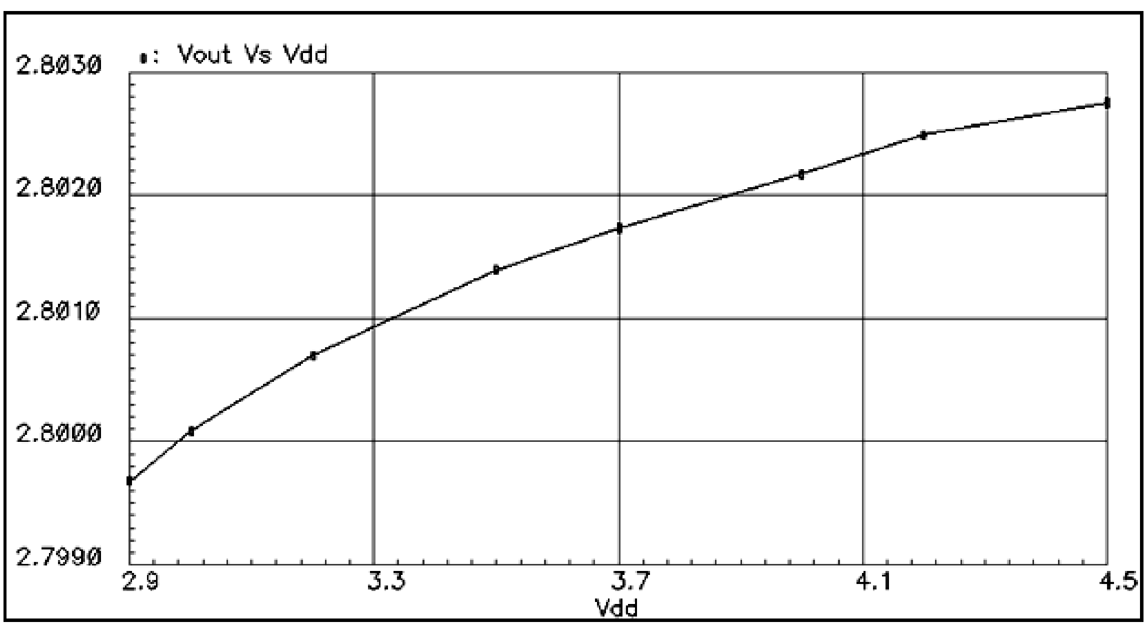


Fig36. Worst Case Line Regulation

TEMPERATURE COEFFICIENT

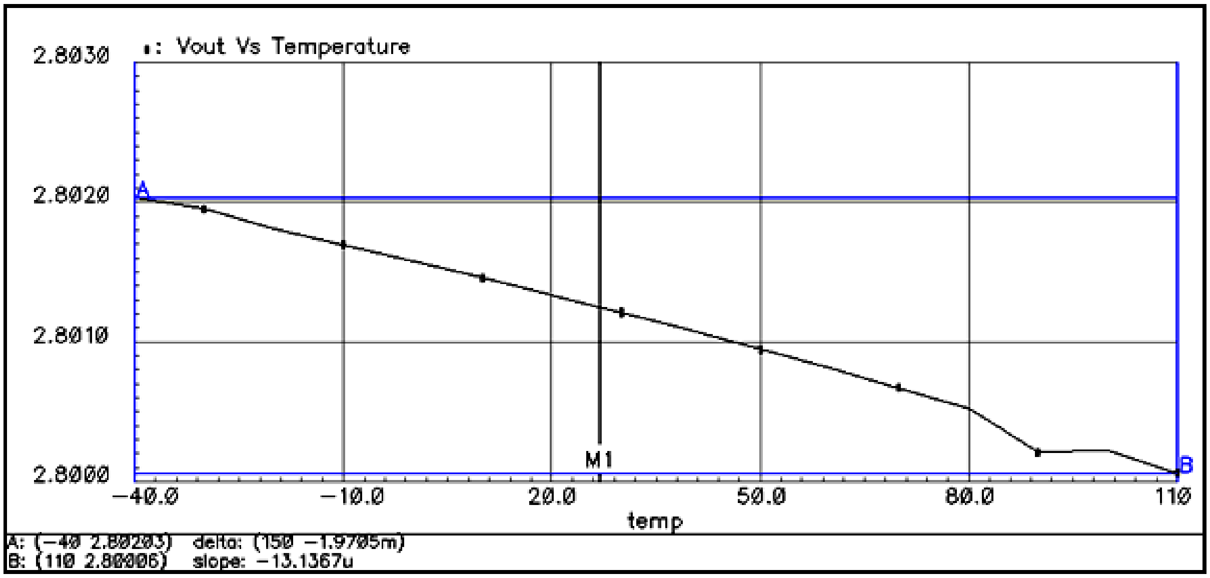


Fig37. Worst Case Temperature Coefficient

Temperature Coefficient = $\frac{\left(V_{ref}(T_2)-V_{ref}(T_1)\right)}{V_{ref}(27°C)*(T_2-T_1)}$ in ppm

$V_{ref}(27°C) = 2.80124$

Hence from the curve Temperature Coefficient = 4.68 ppm

SUMMARY OF RESULTS

Parameter	Typical Values (Normal)	Corner Values (Worst Case)
V_{DD}/V_{OUT}	3 V/2.8 V	3 V/2.8 V
Quiescent Current	$38\ \mu A\ to\ 45\ \mu A$	$30\mu A\ to\ 34\ \mu A$
PSR @ 10 MHz	$-44.35\ dB\ to - 30\ dB$	$-46.3\ dB\ to - 30\ dB$
PSR @ 100 MHz	$-37.5\ dB\ to - 27.8\ dB$	$-37.74\ dB\ to - 27.84$
Output Noise @ 100 Hz	$133.9 \frac{nV}{\sqrt{Hz}}$	$158.5 \frac{nV}{\sqrt{Hz}}$
Output Noise @ 100 KHz	$332 \frac{nV}{\sqrt{Hz}}$	$292 \frac{nV}{\sqrt{Hz}}$
Load Regulation	$0.23 \frac{mV}{mA}$	$0.15 \frac{mV}{mA}$
Line Regulation	$1.168 \frac{mV}{V}$	$1.875 \frac{mV}{V}$
Temperature Coefficient	$4.55\ ppm$	$4.68\ ppm$
Cout/ESR/Lp	$1\mu F/500m\Omega/500pH$	-
I_{Load}	$0 - 20mA$	$0 - 20mA$
On-Chip Capacitor	$0\ F$	$0\ F$
$V_{OUT}\ Ripple$	$< 13mV\ (0 - 2mA)$ $< 27\ mV\ (0 - 20mA)$	$< 8\ mV\ (0 - 2mA)$ $< 14\ mV\ (0 - 20mA)$
Settling Time	$< 15\mu s\ (2mA)$ $< 5\mu s\ (20mA)$	$< 26\ \mu s\ (2mA)$ $< 6\ \mu s\ (20mA)$
DC Gain	$60\ dB - 43\ dB$	$60\ dB - 47\ dB$
Phase Margin	$32° - 84°$	$29.5° - 57°$
*[DC PSR]-[PSR @ 100 MHz]	$10dB - 13\ dB$	$10\ dB - 13dB$

Table3. Performance Summary of LDO

**The difference between the DC PSR and PSR at 100 MHz is maximum of 13 dB. This we can't achieve in internally compensated and conventional LDOs in technologies like 0.6um*

PERFORMANCE COMPARISON

Parameters	Rincon Mora et al. [10]	Liang-Guo Shen et al. [15]	Vadim Ivanov [8]	This Work
Technology	$0.6\mu m$	$0.18\mu m$	$65nm$	$0.6\mu m$
[DC PSR]-[PSR @ 100 MHz]	$30\ dB - 40\ dB$	-	$> 20\ dB$	$10\ dB - 13\ dB$
PSR @ 10 MHz for 5 mA Load	$-30\ dB$	$< -60\ dB$	$> -25\ dB$	$-38.5\ dB$
Load Current	0 – 5mA	0-100mA	0-30mA	0-20mA
$*\ C_{Load}/ESR\ /L_P$	$10pF/0\Omega/0H$	$10\mu F/10m\Omega/\mathbf{0H}$	$> 100pF/0\Omega/0H$	$1\mu F/500m\Omega/500pH$
On-chip capacitance	$60pF$	$1pF$	$> 1pF$	$0\ F$
V_{DD}/V_{OUT}	1.8 V/1.2 V	1.2 V/1 V	$NA/1.37V$	3V/2.8V
$\Delta Vout$	$937\ mV$	$14mV$	$22mV$	$< 27\ mV$
I_Q	$\gg 40\mu A$	$20.5\mu A$	$1\ mA$	$< 50\mu A$
Noise @ 100 Hz	-	-	$700\frac{nV}{\sqrt{Hz}}$	$158.5\frac{nV}{\sqrt{Hz}}$
Power Efficiency	$< 67.2\%$	83.3%	$< 87\%$	93%

Table4. LDO Comparison with other Works

*In reference [15] an ideal off-chip capacitor is assumed (without inductor) which is not practically feasible.

ADDITIONAL WORK

If the load regulation is relaxed, say if the maximum load current is less than 2 mA the following PSR curve can be achieved. $PSR\ @\ 10\ MHz = -55.8\ dB$

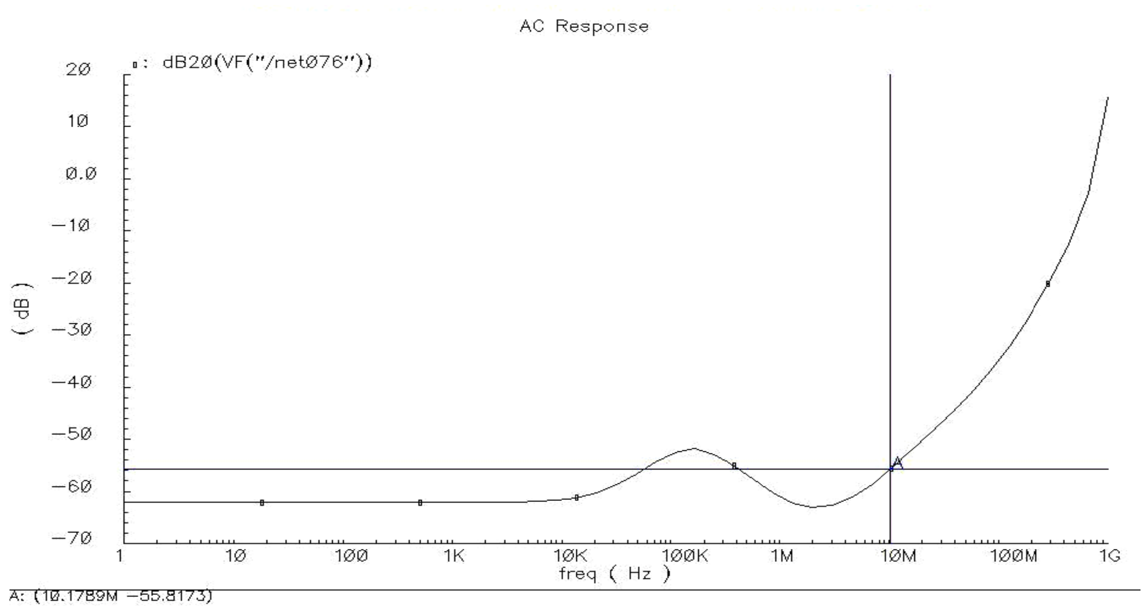


Fig38. PSR of Low Load Current LDO

The reason for achieving good PSR over wide range of frequencies is that the non-dominant poles in the loop are at higher frequencies compared to GBW. But when we need wide load current support the gain need to be increased to regulate which in turn increases the non-dominant frequencies. That's the reason for getting slightly degraded PSR performance in the 20 mA LDO.

FUTURE WORK

1. Additional current can be burnt in the LDO and the non-dominant poles can be moved still to higher frequencies.
2. Gain of the first stage can be boosted so that the DC gain for maximum load current can be increased which will result in better PSR performance.

PHASE NOISE PERFORMANCE WITH INTERNALLY COMPENSATED LDO

The Conventional LDO with lesser PSR BW was used to test the supply rejection capability in the NMOS cross coupled VCO. To emulate the supply noise, resistors were used as a noisy element and a band pass filter was used to inject noise over a little BW around 1MHz, 10MHz and 100MHz, and it's Phase Noise at 1 MHz, 10MHz and 100MHz offsets were observed. The Phase Noise curves are shown below.

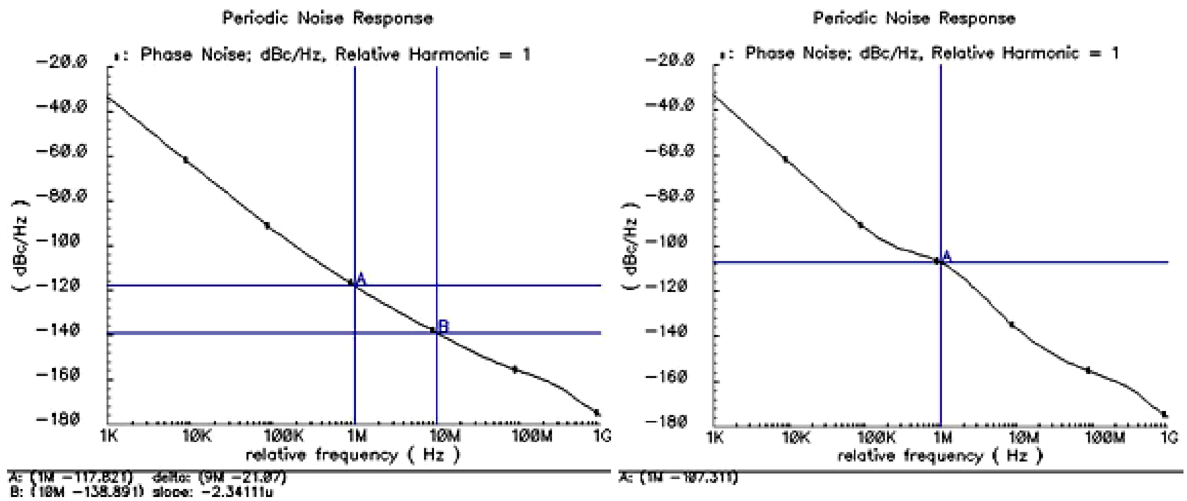


Figure39: Phase Noise for Pure V_{DD} and for noisy V_{DD} with noise around 1MHz

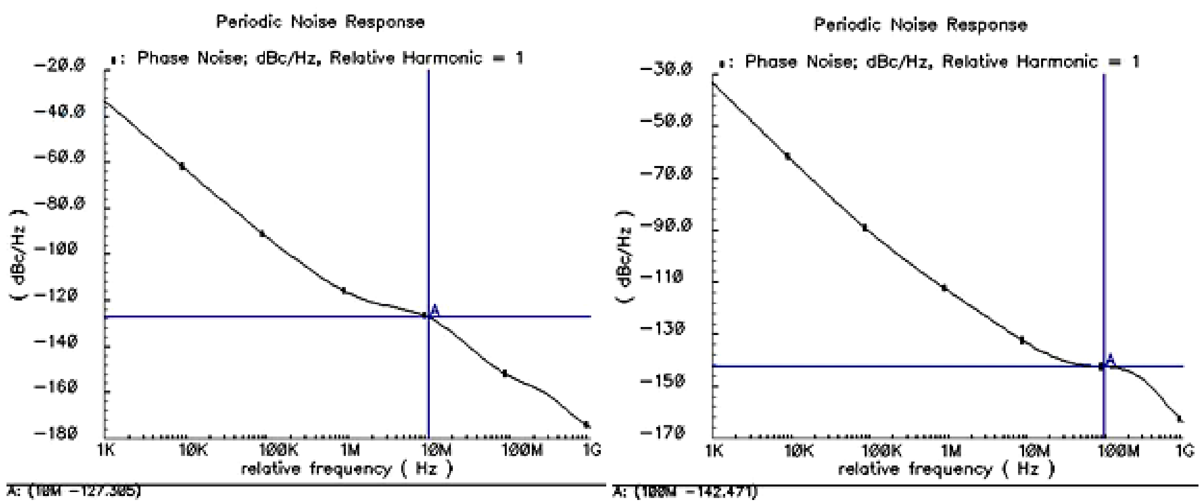


Figure40: Phase Noise for noisy V_{DD} with noise around 10MHz and 100MHz

From these plots it is very well evident there is a definite up-conversion and down-conversion of noise @ Δf around $n\omega_0$ to ω_0.

When noise were concentrated around 1MHz, the phase noise at an offset of 1MHz were affected. Similarly when noise were concentrated around 10MHz (100MHz), the phase noise at an offset of 10MHz(100MHz) were affected. From the plots shown above it can be concluded that the PSR bandwidth is so low that noise at 1MHz and 10MHz tend to affect the phase noise rigourlessly.

PHASE NOISE PERFORMANCE WITH PROPOSED LDO

The Proposed LDO has more or less the same DC PSR, but it has higher PSR band width which very much preserves the phase noise from degradation due to power supply noise. The same test bench was used to model the power supply noise around specific frequency bands, and its effect on Phase noise in the presence of this new LDO is shown below.

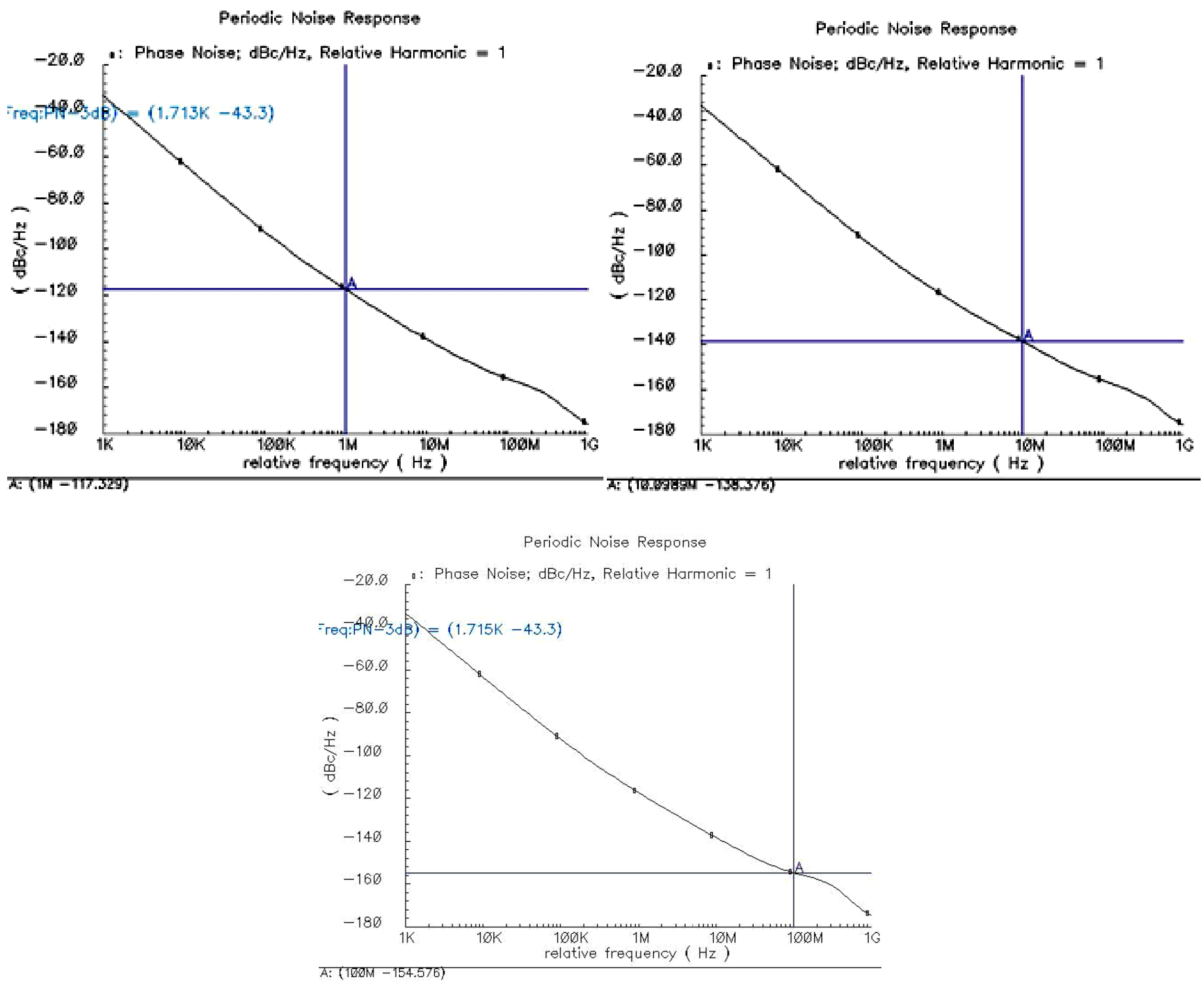


Figure41: Phase Noise for noisy V_{DD} with noise around 1MHz, 10MHz and 100MHz

	Phase Noise in dBc/Hz		
Frequency Offset	**Noise Less V_{DD}**	**Noisy V_{DD} with noise around offset frequency**	**Noisy V_{DD} with noise around offset frequency**
		Internally Compensated LDO	**Proposed LDO**
1MHz	-117.8	-107.3	-117.3
10MHz	-138.8	-127.3	-138.3
100MHz	-156	-142.5	-154.6

Table5: Phase Noise enhanced by using proposed LDO

LAYOUT

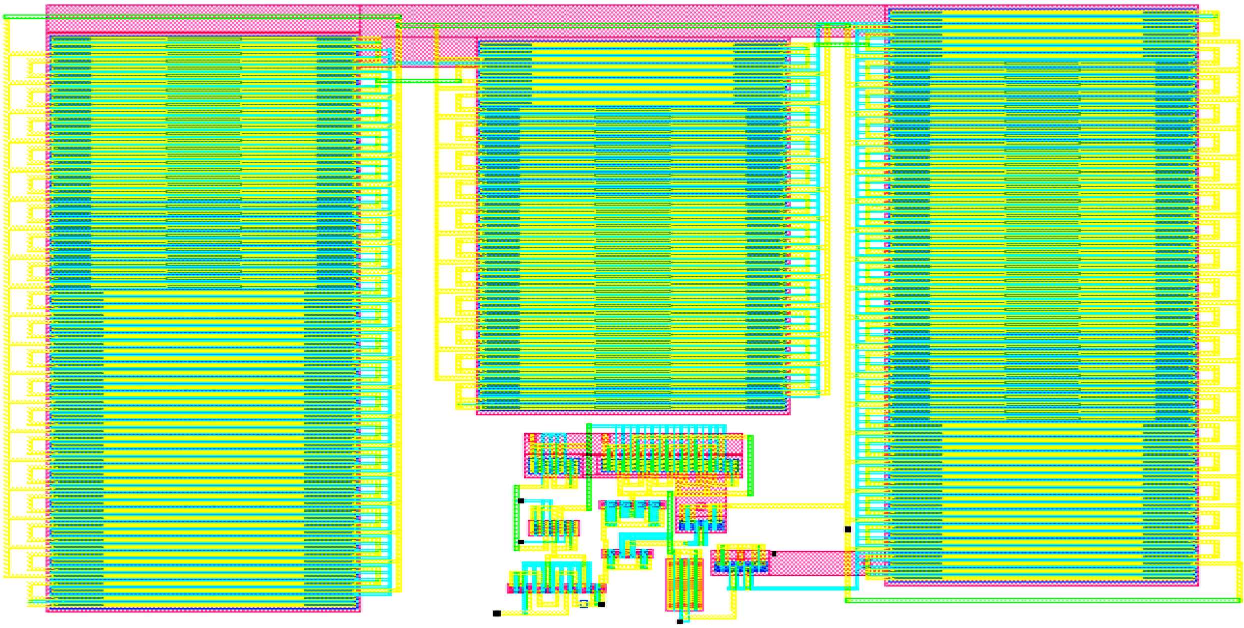


Figure42: Layout of the LDO

LVS RESULT

@(#)$CDS: LVS.exe version 5.1.0 06/20/2007 02:37 (cicsun11) $

Command line: /baby/cadence/ic5141usr5/tools.sun4v/dfII/bin/32bit/LVS.exe -dir /home11/kshiram/cadence/LVS -l -s -t /home11/kshiram/cadence/LVS/layout /home11/kshiram/cadence/LVS/schematic
Like matching is enabled.
Net swapping is enabled.
Using terminal names as correspondence points.
Compiling Diva LVS rules...

Net-list summary for /home11/kshiram/cadence/LVS/layout/netlist

count	
13	nets
7	terminals
240	pmos
28	nmos

Net-list summary for /home11/kshiram/cadence/LVS/schematic/netlist

count	
13	nets
7	terminals
11	pmos
12	nmos

Terminal correspondence points

N1	*N0*	*IB*
N3	*N1*	*VB*
N8	*N8*	*VOUT*
N12	*N11*	*Vin1+*
N6	*N6*	*Vin1-*
N2	*N4*	*gnd*
N11	*N10*	*vdd*

Devices in the rules but not in the netlist:
cap nfet pfet nmos4 pmos4

The net-lists match.

	layout	*schematic*
	instances	
un-matched	*0*	*0*
rewired	*0*	*0*
size errors	*0*	*0*
pruned	*0*	*0*
active	*268*	*23*
total	*268*	*23*
	nets	
un-matched	*0*	*0*
merged	*1*	*0*
pruned	*0*	*0*
active	*13*	*13*
total	*13*	*13*
	terminals	
un-matched	*0*	*0*
matched but different type	*0*	*0*
total	*7*	*7*

Probe files from /home11/kshiram/cadence/LVS/schematic
devbad.out:
netbad.out:
mergenet.out:
termbad.out:
prunenet.out:
prunedev.out:
audit.out:

Probe files from /home11/kshiram/cadence/LVS/layout
devbad.out:
netbad.out:
mergenet.out:
termbad.out:
prunenet.out:
prunedev.out:
audit.out:

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